

Floyd–Hoare Logic for Quantum Programs

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Floyd–Hoare logic is a foundation of axiomatic semantics of classical programs, and it provides effective proof techniques for reasoning about correctness of classical programs. To offer similar techniques for quantum program verification and to build a logical foundation of programming methodology for quantum computers, we develop a full-fledged Floyd–Hoare logic for both partial and total correctness of quantum programs. It is proved that this logic is (relatively) complete by exploiting the power of weakest preconditions and weakest liberal preconditions for quantum programs.

Categories and Subject Descriptors: D.3.1 [Programming Languages]: Formal Definitions and Theory; F.3.1 [Logics and Meanings of Programs]: Specifying and Verifying and Reasoning about Programs—*Specification techniques*; F.3.2 [Logics and Meanings of Programs]: Semantics of Programming Languages—*Operational semantics*

General Terms: Theory, Languages

Additional Key Words and Phrases: Quantum computation, programming language, axiomatic semantics, Floyd–Hoare logic, completeness

ACM Reference Format:

Ying, M. 2011. Floyd–Hoare logic for quantum programs. *ACM Trans. Program. Lang. Syst.* 33, 6, Article 19 (December 2011), 49 pages.

DOI = 10.1145/2049706.2049708 <http://doi.acm.org/10.1145/2049706.2049708>

1. INTRODUCTION

Even though quantum hardware is still in its infancy, people widely believe that building a large-scale and functional quantum computer is merely a matter of time and concentrated effort. The history of classical computing arouses hope that once quantum computers come into being, quantum programming languages and quantum software development techniques will play a key role in exploiting the power of quantum computers. With the expectation of offering effective programming techniques for quantum computers, several quantum programming languages have already been designed in recent years. The earliest proposal for quantum programming language was made by Knill [1996]. The first real quantum programming language, QCL, was proposed by Ömer [2003]; he also implemented a simulator for this language. A quantum programming language in the style of Dijkstra’s guarded-command language, qGCL, was presented by Sanders and Zuliani [2000]. A quantum extension of C++ was proposed by Bettelli et al. [2003], and it was implemented in the form of a C++ library. The first and

This work was partly supported by the National Natural Science Foundation of China, Grant 60736011 and the Australian Research Council, Grant DP110103473.

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DOI 10.1145/2049706.2049708 <http://doi.acm.org/10.1145/2049706.2049708>

very influential quantum language of the functional programming paradigm, QPL, was defined by Selinger [2004b] based on the idea of classical control and quantum data. Altenkirch and Grattage [2005] defined another functional programming language for quantum computing, QML, in which both control and data may be quantum. For an excellent survey of quantum programming languages, see Selinger [2004a] and Gay [2006].

The fact that human intuition is much better adapted to the classical world than to the quantum world is one of the major reasons why it is difficult to find efficient quantum algorithms. It also implies that programmers will commit many more faults in designing programs for quantum computers than programming classical computers. Thus, it is even more critical to give clear and formal semantics to quantum programming languages and provide formal methods for reasoning about quantum programs. Indeed, various semantic approaches to quantum programs have been proposed in recent literature. For example, an operational semantics was given to Sanders and Zuliani's language, qGCL [Sanders and Zuliani 2000] by treating an observation (quantum measurement) procedure as a probabilistic choice; a denotational semantics was defined for Selinger's language, QPL [Selinger 2004b] by interpreting quantum programs as super-operators; and the denotational semantics of Altenkirch and Grattage's language, QML [Altenkirch and Grattage 2005] was described in category-theoretic terms. In addition, a language-independent approach to the semantics of quantum programs was proposed by D'Hondt and Panangaden [2006], who introduced an intrinsic notion of quantum weakest precondition and established a beautiful Stone-type duality between state transition semantics and predicate transformer semantics for quantum programs. Quantum predicate transformer semantics was also systematically studied by Ying et al. [2010].

As to proof systems for reasoning about quantum programs, Baltag and Smets [2004, 2006] and Akatov [2005] presented a dynamic logic formalism for information flows in quantum systems, that is capable of describing various quantum operations such as unitary evolution and quantum measurement, and particularly entanglements in multipartite quantum systems. Brunet and Jorrand [2004] introduced a way of applying Birkhoff and von Neumann's quantum logic [1936] to the study of quantum programs by expanding the usual propositional languages with new primitives representing unitary transformations and quantum measurements. Chadha et al. [2006] proposed a Floyd–Hoare-style proof system for reasoning about imperative quantum programs using a quantitative state logic, but only bounded iterations are allowed in their programming language. Feng et al. [2007] found some useful proof rules for reasoning about quantum loops, generalizing some of Morgan's proof rules for probabilistic loops [Morgan 1995]. To the best of the present author's knowledge, however, no complete Floyd–Hoare logic for quantum programs has been reported in the literature.

The main contribution of the present article is the establishment of a full-fledged Floyd–Hoare logic for deterministic quantum programs based on Selinger's idea of modeling quantum programs as super-operators, and on D'Hondt and Panangaden's notion of a quantum predicate as an Hermitian operator [D'Hondt and Panangaden 2006]. This logic includes a proof system for partial correctness and a proof system for total correctness for deterministic quantum programs. In particular, we are able to prove its (relative) completeness by exploiting the power of weakest preconditions and weakest liberal preconditions for quantum programs.

The article is organized as follows: For the convenience of the reader, we recall some basic concepts of Hilbert spaces, as well as the fundamental postulates of quantum mechanics, in Section 2. Another aim of Section 2 is to fix the notation used in the following sections. In Section 3, we define the syntax of deterministic quantum programs, about which the Floyd–Hoare logic in this article is designed to reason. Such quantum

programs are quantum extensions of classical **while**-programs (cf., Apt and Olderog [1997], Ch. 3). In Section 4, an operational semantics of quantum programs is given in terms of transitions between quantum configurations, which consist of a quantum program still to be executed and a (partial) density operator expressing the current state of program variables. Hence, in Section 5, we introduce a denotational semantics of quantum programs based on the operational semantics. A denotational semantics of a quantum program is defined as a function from partial density operators to themselves. In Section 6, we adopt D’Hondt and Panangaden’s definition of quantum predicates as Hermitian operators. Then, a correctness formula is defined to be a quantum extension of a Hoare triple, which consists of two quantum predicates, namely precondition and postcondition, as well as a quantum program. Furthermore, the notions of partial and total correctness can be introduced for quantum programs using their denotational semantics. In Section 7, the weakest precondition and weakest liberal precondition for quantum programs are defined in terms of total and partial correctness, respectively, in the usual way. With the long preparation of the previous sections, the Floyd–Hoare logic for quantum programs is finally established in Sections 8 and 9. In Section 8, a proof system for partial correctness of quantum programs is presented, and its (relative) completeness is proved; in Section 9, after introducing the notion of a bound function for quantum loops, a proof system for total correctness of quantum programs is given, and its (relative) completeness is also proved. As an illustrative example, the quantum Floyd–Hoare logic in Sections 8 and 9 is used in Section 10 to prove the correctness of the Grover algorithm [Grover 1996]. A brief conclusion is drawn, and some open problems for further studies are pointed out in Section 11.

2. PRELIMINARIES

2.1. Hilbert Spaces

We write $\mathbb{C}$ for the set of complex numbers. For each complex number $\lambda \in \mathbb{C}$, λ^* stands for the conjugate of λ . We adopt the Dirac notation: $|\varphi\rangle$, $|\psi\rangle$, which stand for vectors. A (complex) vector space is a nonempty set $\mathcal{H}$ together with two operations: vector addition $+$: $\mathcal{H} \times \mathcal{H} \rightarrow \mathcal{H}$ and scalar multiplication $\cdot$: $\mathbb{C} \times \mathcal{H} \rightarrow \mathcal{H}$, satisfying the following conditions:

- (1) $(\mathcal{H}, +)$ is an Abelian group; its zero element 0 is called the zero vector;
- (2) $1|\varphi\rangle = |\varphi\rangle$;
- (3) $\lambda(\mu|\varphi\rangle) = \lambda\mu|\varphi\rangle$;
- (4) $(\lambda + \mu)|\varphi\rangle = \lambda|\varphi\rangle + \mu|\varphi\rangle$; and
- (5) $\lambda(|\varphi\rangle + |\psi\rangle) = \lambda|\varphi\rangle + \lambda|\psi\rangle$

for any $\lambda, \mu \in \mathbb{C}$ and for any vectors $|\varphi\rangle, |\psi\rangle \in \mathcal{H}$.

An inner product over a vector space $\mathcal{H}$ is a mapping $\langle \cdot | \cdot \rangle : \mathcal{H} \times \mathcal{H} \rightarrow \mathbb{C}$ satisfying the following properties:

- (1) $\langle \varphi | \varphi \rangle \geq 0$ with equality if and only if $|\varphi\rangle = 0$;
- (2) $\langle \varphi | \psi \rangle = \langle \psi | \varphi \rangle^*$; and
- (3) $\langle \varphi | \lambda_1 \psi_1 + \lambda_2 \psi_2 \rangle = \lambda_1 \langle \varphi | \psi_1 \rangle + \lambda_2 \langle \varphi | \psi_2 \rangle$

for any $|\varphi\rangle, |\psi\rangle, |\psi_1\rangle, |\psi_2\rangle \in \mathcal{H}$ and for any $\lambda_1, \lambda_2 \in \mathbb{C}$. Sometimes, we also write $(|\varphi\rangle, |\psi\rangle)$ for the inner product $\langle \varphi | \psi \rangle$ of vectors $|\varphi\rangle$ and $|\psi\rangle$.

For any vector $|\psi\rangle$ in $\mathcal{H}$, its length $\| \psi \|$ is defined as $\sqrt{\langle \psi | \psi \rangle}$. A vector $|\psi\rangle$ is called a unit vector if $\| \psi \| = 1$. We say that two vectors $|\varphi\rangle$ and $|\psi\rangle$ are orthogonal, written $|\varphi\rangle \perp |\psi\rangle$, if $\langle \varphi | \psi \rangle = 0$. Let $\{|\psi_n\rangle\}$ be a sequence of vectors in $\mathcal{H}$ and $|\psi\rangle \in \mathcal{H}$. If for any $\epsilon > 0$, there exists a positive integer N such that $\| \psi_m - \psi_n \| < \epsilon$ for all $m, n \geq N$, then $\{|\psi_n\rangle\}$ is called a Cauchy sequence. If for any $\epsilon > 0$, there exists a positive integer N

such that $\|\psi_n - \psi\| < \epsilon$ for all $n \geq N$, then $|\psi\rangle$ is called a limit of $\{|\psi_n\rangle\}$, and we write $|\psi\rangle = \lim_{n \rightarrow \infty} |\psi_n\rangle$.

A family $\{|\psi_i\rangle\}_{i \in I}$ of vectors in $\mathcal{H}$ is said to be summable with the sum $|\psi\rangle$, and we write $|\psi\rangle = \sum_{i \in I} |\psi_i\rangle$ if for any $\epsilon > 0$ there is a finite subset J of I such that

$$\left\| \psi - \sum_{i \in K} \psi_i \right\| < \epsilon$$

for every finite subset K of I containing J . A family $\{|\psi_i\rangle\}_{i \in I}$ of unit vectors is called an orthonormal basis of $\mathcal{H}$ if

- (1) $|\psi_i\rangle \perp |\psi_j\rangle$ for any $i, j \in I$ with $i \neq j$; and
- (2) $|\psi\rangle = \sum_{i \in I} \langle \psi_i | \psi \rangle |\psi_i\rangle$ for each $|\psi\rangle \in \mathcal{H}$.

In this case, the cardinality of I is called the dimension of $\mathcal{H}$.

A Hilbert space is defined as a complete inner product space; that is, an inner product space in which each Cauchy sequence of vectors has a limit. According to a basic postulate of quantum mechanics, the state space of an isolated quantum system is represented by a Hilbert space, and a pure state of the system is described by a unit vector (up to a phase shift) in its state space.

Example 2.1.

- (1) The state space of qubits is the 2-dimensional Hilbert space:

$$\mathcal{H}_2 = \{\alpha|0\rangle + \beta|1\rangle : \alpha, \beta \in \mathbb{C}\}.$$

The inner product in $\mathcal{H}_2$ is defined by

$$(\alpha|0\rangle + \beta|1\rangle, \alpha'|0\rangle + \beta'|1\rangle) = \alpha^* \alpha' + \beta^* \beta'$$

for all $\alpha, \alpha', \beta, \beta' \in \mathbb{C}$. Then $\{|0\rangle, |1\rangle\}$ is an orthonormal basis of $\mathcal{H}_2$, called the computational basis. For any $\alpha, \beta \in \mathbb{C}$, the vector $\alpha|0\rangle + \beta|1\rangle$ in $\mathcal{H}_2$ is often represented by a column vector

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

in $\mathbb{C}^2$; in particular,

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

The following two states of a qubit will often be considered in the sequent examples:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle).$$

- (2) The space l_2 of square summable sequences is

$$\mathcal{H}_\infty = \left\{ \sum_{n=-\infty}^{\infty} \alpha_n |n\rangle : \alpha_n \in \mathbb{C} \text{ for all } n \in \mathbb{Z} \text{ and } \sum_{n=-\infty}^{\infty} |\alpha_n|^2 < \infty \right\},$$

where $\mathbb{Z}$ is the set of integers. The inner product in $\mathcal{H}_\infty$ is defined by

$$\left(\sum_{n=-\infty}^{\infty} \alpha_n |n\rangle, \sum_{n=-\infty}^{\infty} \alpha'_n |n\rangle \right) = \sum_{n=-\infty}^{\infty} \alpha_n^* \alpha'_n$$

for all $\alpha_n, \alpha'_n \in \mathbb{C}$, $-\infty < n < \infty$. Then $\{|n\rangle : n \in \mathbb{Z}\}$ is an orthonormal basis of $\mathcal{H}_\infty$, called the computational basis.

A (linear) operator on a Hilbert space $\mathcal{H}$ is a mapping $A : \mathcal{H} \rightarrow \mathcal{H}$ satisfying the following conditions:

- (1) $A(|\varphi\rangle + |\psi\rangle) = A|\varphi\rangle + A|\psi\rangle$;
- (2) $A(\lambda|\psi\rangle) = \lambda A|\psi\rangle$

for all $|\varphi\rangle, |\psi\rangle \in \mathcal{H}$ and $\lambda \in \mathbb{C}$. If $\{|\psi_i\rangle\}$ is an orthonormal basis of $\mathcal{H}$, then an operator A is uniquely determined by the images $\{A|\psi_i\rangle\}$ of basis vectors $\{|\psi_i\rangle\}$ under A . In particular, A can be represented by a matrix

$$A = (\langle\psi_i|A|\psi_j\rangle)_{ij}$$

when $\mathcal{H}$ is finite-dimensional.

An operator A on $\mathcal{H}$ is said to be bounded if there is a constant $C \geq 0$ such that $\|A|\psi\rangle\| \leq C \cdot \|\psi\|$ for all $|\psi\rangle \in \mathcal{H}$. In this article, we only consider bounded operators. We write $\mathcal{L}(\mathcal{H})$ for the set of bounded operators on $\mathcal{H}$. The identity operator on $\mathcal{H}$ is denoted $I_{\mathcal{H}}$; that is, $I_{\mathcal{H}}|\psi\rangle = |\psi\rangle$ for all $|\psi\rangle \in \mathcal{H}$, and the zero operator on $\mathcal{H}$ that maps every vector in $\mathcal{H}$ to the zero vector is denoted $0_{\mathcal{H}}$.

For any operator A on $\mathcal{H}$, there exists a unique linear operator $A^\dagger$ on $\mathcal{H}$ such that

$$(|\varphi\rangle, A|\psi\rangle) = (A^\dagger|\varphi\rangle, |\psi\rangle)$$

for all $|\varphi\rangle, |\psi\rangle \in \mathcal{H}$. The operator $A^\dagger$ is called the adjoint of A . An operator M on $\mathcal{H}$ is said to be Hermitian if $M^\dagger = M$. An operator A on $\mathcal{H}$ is said to be positive if $\langle\psi|A|\psi\rangle \geq 0$ for all states $|\psi\rangle \in \mathcal{H}$. It is easy to see that every positive operator is Hermitian.

An operator A is said to be a trace class operator if $\{\langle\psi_i|A|\psi_i\rangle\}_{i \in I}$ is summable for any orthonormal basis $\{|\psi_i\rangle\}_{i \in I}$ of $\mathcal{H}$; in this case, the trace $tr(A)$ of A is defined as

$$tr(A) = \sum_i \langle\psi_i|A|\psi_i\rangle$$

where $\{|\psi_i\rangle\}$ is an orthonormal basis of $\mathcal{H}$. It can be shown that $tr(A)$ is independent of the choice of $\{|\psi_i\rangle\}$.

A density operator ρ on a Hilbert space $\mathcal{H}$ is defined to be a positive operator with $tr(\rho) = 1$. Then a mixed state of a quantum system with state space $\mathcal{H}$ is described by a density operator on $\mathcal{H}$. In this article, we take a slightly generalized notion of the density operator: a partial density operator ρ is a positive operator with $tr(\rho) \leq 1$. In particular, the zero operator is a partial density operator. The set of partial density operators is denoted $\mathcal{D}^-(\mathcal{H})$. A partial density operator can also be defined by an ensemble of pure states. Suppose that a quantum system is in one of a number of pure states $|\psi_i\rangle$, with respective probabilities p_i , where it is required that $\sum_i p_i \leq 1$. Then

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$$

is a partial density operator. Conversely, any partial density operator can be generated in such a way. The notion of partial density operator allows us to follow Selinger's normalization convention ([Selinger 2004b], Convention 3.3), which is very convenient in the presentation of the semantics of quantum programs.

We can define a partial order between operators, called the Löwner partial order: for any $A, B \in \mathcal{L}(\mathcal{H})$, $A \sqsubseteq B$ if $B - A$ is a positive operator.

LEMMA 2.1. $A \sqsubseteq B$ if and only if $tr(A\rho) \leq tr(B\rho)$ for all (partial) density operators ρ .

2.2. Unitary Transformations

An operator U on $\mathcal{H}$ is called a unitary transformation if the adjoint of U is its inverse; that is, $U^\dagger U = I_{\mathcal{H}}$, where $I_{\mathcal{H}}$ is the identity operator on $\mathcal{H}$.

Example 2.2.

- (1) The most frequently used unitary operators on qubits are the Hadamard transformation:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

and the Pauli matrices:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

- (2) Let k be an integer. Then the k -translation operator on $\mathcal{H}_\infty$ is defined by

$$U_{+k}|n\rangle = |n+k\rangle$$

for all $n \in \mathbb{Z}$. It is easy to verify that U_{+k} is a unitary operator.

The basic postulate of quantum mechanics about evolution of systems may be stated as follows: suppose that the states of a closed quantum system at times t_0 and t are $|\psi_0\rangle$ and $|\psi\rangle$, respectively. Then they are related to each other by a unitary operator U which depends only on the times t_0 and t ,

$$|\psi\rangle = U|\psi_0\rangle.$$

This postulate can be reformulated in the language of density operators, as follows: The state ρ of a closed quantum system at time t is related to its state ρ_0 at time t_0 by a unitary operator U which depends only on the times t and t_0 ,

$$\rho = U\rho_0U^\dagger.$$

2.3. Quantum Measurements

A quantum measurement on a system with state space $\mathcal{H}$ is described by a collection $\{M_m\}$ of operators on $\mathcal{H}$ satisfying the following normalization condition:

$$\sum_m M_m^\dagger M_m = I_{\mathcal{H}},$$

where M_m are called measurement operators, and the index m stands for the measurement outcomes that may occur in the experiment. If the state of a quantum system is a pure state $|\psi\rangle$ immediately before the measurement, then the probability that result m occurs is

$$p(m) = \langle \psi | M_m^\dagger M_m | \psi \rangle$$

and the state of the system after the measurement is

$$|\psi_m\rangle = \frac{M_m|\psi\rangle}{\sqrt{p(m)}}.$$

We can also formulate the quantum measurement postulate in the language of density operators. If the state of a quantum system is ρ immediately before measurement $\{M_m\}$ is performed on it, then the probability that result m occur is

$$p(m) = \text{tr}(M_m^\dagger M_m \rho),$$

and the state of the system after the measurement is

$$\rho_m = \frac{M_m \rho M_m^\dagger}{p(m)}.$$

A special class of quantum measurements will be used frequently in the sequel. If a measurement M has only two outcomes, say 0 and 1; that is, $M = \{M_0, M_1\}$, then we often call M a yes-no measurement, with 0 corresponding to “no” and 1 to “yes”.

2.4. Tensor Products of Hilbert Spaces

The state space of a composite quantum system is the tensor product of the state spaces of its components. In this section, we recall the definition of the tensor product of a family $\{\mathcal{H}_i\}$ of Hilbert spaces. For simplicity of presentation, it is assumed that the set of quantum variables is countably infinite, and the type of each quantum variable is either **Boolean** or **integer** (see the next section). Thus, we only need to consider a finite or countably infinite family $\{\mathcal{H}_i\}$ where each $\mathcal{H}_i$ is finite-dimensional or countably infinite-dimensional. A more general notion of tensor product introduced by von Neumann [1938] should be adopted in order to generalize the results of this article to the case of more quantum variables and other data types.

Let $\{|\psi_{i,j_i}\rangle\}$ be an orthonormal basis of $\mathcal{H}_i$ for each i . We write $\mathcal{B}$ for the set of tensor products of basis vectors of all $\mathcal{H}_i$; that is,

$$\mathcal{B} = \left\{ \bigotimes_i |\psi_{i,j_i}\rangle \right\}.$$

Then $\mathcal{B}$ is a finite or countably infinite set. We only consider the case that $\mathcal{B}$ is infinite, since the case of finite $\mathcal{B}$ is simpler. Now, $\mathcal{B}$ can be written in the form of a sequence of vectors:

$$\mathcal{B} = \{|\varphi_n\rangle : n = 0, 1, \dots\}.$$

The tensor product of $\{\mathcal{H}_i\}$ is defined to be the Hilbert space spanned by $\mathcal{B}$, that is,

$$\bigotimes_i \mathcal{H}_i = \left\{ \sum_n |\varphi_n\rangle : \alpha_n \in \mathbb{C} \text{ for all } n \geq 0 \text{ and } \sum_n |\alpha_n|^2 < \infty \right\}.$$

We define the inner product in $\bigotimes_i \mathcal{H}_i$ as follows:

$$\left(\sum_n \alpha_n |\varphi_n\rangle, \sum_n \alpha'_n |\varphi_n\rangle \right) = \sum_n \alpha_n^* \alpha'_n$$

for any $\alpha_n, \alpha'_n \in \mathbb{C}$, $n \geq 0$. It is easy to see that $\bigotimes_i \mathcal{H}_i$ is isomorphic to the space $\mathcal{H}_\infty$ of square summable sequences defined in Example 2.1(2).

The notion of partial trace is very useful for the description of a subsystem of a composite quantum system. Let $\mathcal{H}$ and $\mathcal{K}$ be two Hilbert spaces and operator $A \in \mathcal{L}(\mathcal{H} \otimes \mathcal{K})$. Then the partial trace of A on $\mathcal{H}$ is defined as

$$tr_{\mathcal{K}}(A) = \sum_i (I_{\mathcal{H}} \otimes \langle \psi_i |) A (I_{\mathcal{H}} \otimes | \psi_i \rangle),$$

which is an operator on $\mathcal{H}$, where $\{|\psi_i\rangle\}$ is an orthonormal basis of $\mathcal{K}$. It can be shown that $tr_{\mathcal{K}}(A)$ does not depend on the choice of $\{|\psi_i\rangle\}$. In particular, if $\mathcal{H}_1$ and $\mathcal{H}_2$ are the state spaces of quantum systems q_1 and q_2 , respectively, and the state of their composite system $q_1 q_2$ is described by a density operator $\rho \in \mathcal{D}^-(\mathcal{H}_1 \otimes \mathcal{H}_2)$, then $tr_{\mathcal{H}_2}(\rho)$ is the description for the state of component system q_1 .

3. SYNTAX OF QUANTUM PROGRAMS

We assume a countably infinite set Var of quantum variables. The symbols $q, q', q'', q_0, q_1, q_2, \dots$ will be used as metavariables ranging over quantum variables. Recall that in classical computation, we use a type to denote the domain of a variable. Thus, in quantum computation, **a type should be the state space of a quantum system denoted by some quantum variable**. Formally, a type t is a name of a Hilbert space $\mathcal{H}_t$. In this article, we only consider two basic types: **Boolean** and **integer**. The results obtained in this article can be easily generalized to the case with more types. The Hilbert spaces denoted by **Boolean** and **integer** are the following:

$$\mathcal{H}_{\text{Boolean}} = \mathcal{H}_2,$$

$$\mathcal{H}_{\text{integer}} = \mathcal{H}_\infty.$$

Note that the sets denoted by types **Boolean** and **integer** in classical computation are exactly the computational bases of $\mathcal{H}_{\text{Boolean}}$ and $\mathcal{H}_{\text{integer}}$, respectively (see Example 2.1). Now we assume that each quantum variable q has a type $type(q)$, which is either **Boolean** or **integer**. The state space $\mathcal{H}_q$ of a quantum variable q is the Hilbert space denoted by its type; that is,

$$\mathcal{H}_q = \mathcal{H}_{type(q)}.$$

A quantum register is defined to be a finite sequence of distinct quantum variables. The state space of a quantum register $\bar{q} = q_1, \dots, q_n$ is the tensor product of the state spaces of the quantum variables occurring in $\bar{q}$; that is,

$$\mathcal{H}_{\bar{q}} = \bigotimes_{i=1}^n \mathcal{H}_{q_i}.$$

Now we are able to define the syntax of quantum programs. The quantum programs considered in this article are quantum extensions of classical **while**-programs. Formally, they are generated by the following grammar:

$$S ::= \textbf{skip} \mid q := 0 \mid \bar{q} := U\bar{q} \mid S_1; S_2 \mid \textbf{measure } M[\bar{q}] : \bar{S} \mid \textbf{while } M[\bar{q}] = 1 \textbf{ do } S$$

where

- q is a quantum variable and $\bar{q}$ a quantum register;
- U in the statement “ $\bar{q} := U\bar{q}$ ” is a unitary operator on $\mathcal{H}_{\bar{q}}$. In particular, if $type(q) = \textbf{integer}$, then the statement $q := U_{+k}q$, where U_{+k} is the k -translation operator, will often be abbreviated to $q := q + k$;
- in the statement “**measure** $M[\bar{q}] : \bar{S}$ ”, $M = \{M_m\}$ is a measurement on the state space $\mathcal{H}_{\bar{q}}$ of $\bar{q}$, and $\bar{S} = \{S_m\}$ is a set of quantum programs such that each outcome m of measurement M corresponds to S_m ;
- $M = \{M_0, M_1\}$ in the statement “**while** $M[\bar{q}] = 1$ **do** S ” is a yes-no measurement on $\mathcal{H}_{\bar{q}}$.

In applications, the unitary operators and measurements in quantum programs are usually chosen from a specific set, which is (approximately) universal; for example, if a program only contains quantum variables of type **Boolean**, then the unitary operators can be taken from the standard set of universal gates, consisting of the Hadamard, phase, CNOT, and $\pi/8$ gates (see Nielsen and Chuang [2000], Ch. 4), and we only need the measurement in the computational basis. For simplicity of the presentation, however, we choose not to specify these unitary operators and measurements because choosing them does make any essential difference in the theory developed in this article.

The intuitive meaning of these quantum program constructs will become clear after introducing their operational semantics in the next section.

We use “ $\equiv$ ” to denote syntactic equality of quantum programs. The following technical definition will be needed in the sequel.

Definition 3.1. The set $\text{var}(S)$ of quantum variables in quantum program S is recursively defined as follows:

- (1) If $S \equiv \mathbf{skip}$, then $\text{var}(S) = \emptyset$;
- (2) If $S \equiv q := 0$, then $\text{var}(S) = \{q\}$;
- (3) If $S \equiv \bar{q} := U\bar{q}$, then $\text{var}(S) = \{\bar{q}\}$;
- (4) If $S \equiv S_1; S_2$, then $\text{var}(S) = \text{var}(S_1) \cup \text{var}(S_2)$;
- (5) If $S \equiv \mathbf{measure} \ M[\bar{q}] : \bar{S}$, then

$$\text{var}(S) = \{\bar{q}\} \cup \bigcup_m \text{var}(S_m);$$

- (6) If $S \equiv \mathbf{while} \ M[\bar{q}] = 1 \ \mathbf{do} \ S$, then $\text{var}(S) = \{\bar{q}\} \cup \text{var}(S)$.

4. OPERATIONAL SEMANTICS OF QUANTUM PROGRAMS

We write $\mathcal{H}_{\text{all}}$ for the tensor product of the state spaces of all quantum variables; that is,

$$\mathcal{H}_{\text{all}} = \bigotimes_{\text{all } q} \mathcal{H}_q.$$

For simplicity of presentation, we use E to denote the empty program. A quantum configuration is a pair $\langle S, \rho \rangle$, where S is a quantum program or E , $\rho \in \mathcal{D}^-(\mathcal{H}_{\text{all}})$ is a partial density operator on $\mathcal{H}_{\text{all}}$, and it is used to indicate the (global) state of quantum variables.

Let $\bar{q} = q_1, \dots, q_n$ be a quantum register. A linear operator A on $\mathcal{H}_{\bar{q}}$ has a cylinder extension

$$A \otimes I_{\text{Var} \setminus \{\bar{q}\}} \tag{1}$$

on $\mathcal{H}_{\text{all}}$, where $I_{\text{Var} \setminus \{\bar{q}\}}$ is the identity operator on the Hilbert space

$$\bigotimes_{q \in \text{var} \setminus \{\bar{q}\}} \mathcal{H}_q.$$

In the sequel, we simply write A for its extension Eq. (1), and it can be easily recognized from the context, without any risk of confusion.

The operational semantics of quantum program is defined as a transition relation $\rightarrow$ between quantum configurations. By a transition

$$\langle S, \rho \rangle \rightarrow \langle S', \rho' \rangle$$

we mean that after executing quantum program S , one step in state ρ , the state of quantum variables becomes ρ' , and S' is the remainder of S still to be executed. In particular, if $S' \equiv E$, then S terminates in state ρ' . The transition relation $\rightarrow$ is given by the transition rules in Figure 1.

The meanings of various program constructs are precisely specified by the transition rules in Figure 1. These rules deserve careful explanations. The statement “**skip**” does nothing and terminates immediately. The initialization “ $q := 0$ ” sets quantum variable q to the basis state $|0\rangle$. In the rule (Initialization) and in the sequel, we use $|\varphi\rangle_q$ to indicate the state $|\varphi\rangle$ of the system denoted by quantum variable q . Thus, for any states $|\varphi\rangle, |\psi\rangle$ of the q system, we can define an operator $|\varphi\rangle_q \langle \psi|$ on the state space of

$$(Skip) \quad \overline{\langle \mathbf{skip}, \rho \rangle \rightarrow \langle E, \rho \rangle}$$

$$(Initialization) \quad \overline{\langle q := 0, \rho \rangle \rightarrow \langle E, \rho_0^q \rangle}$$

where :

$$\rho_0^q = |0\rangle_q \langle 0| \rho |0\rangle_q \langle 0| + |0\rangle_q \langle 1| \rho |1\rangle_q \langle 0|$$

if $type(q) = \mathbf{Boolean}$, and

$$\rho_0^q = \sum_{n=-\infty}^{\infty} |0\rangle_q \langle n| \rho |n\rangle_q \langle 0|$$

if $type(q) = \mathbf{integer}$.

$$(Unitary Transformation) \quad \overline{\langle \bar{q} := U\bar{q}, \rho \rangle \rightarrow \langle E, U\rho U^\dagger \rangle}$$

$$(Sequential Composition) \quad \frac{\langle S_1, \rho \rangle \rightarrow \langle S'_1, \rho' \rangle}{\langle S_1; S_2, \rho \rangle \rightarrow \langle S'_1; S_2, \rho \rangle}$$

where we make the convention that $E; S_2 \equiv S_2$.

$$(Measurement) \quad \overline{\langle \mathbf{measure } M[\bar{q}] : \bar{S}, \rho \rangle \rightarrow \langle S_m, M_m \rho M_m^\dagger \rangle}$$

for each outcome m of measurement $M = \{M_m\}$

$$(Loop 0) \quad \overline{\langle \mathbf{while } M[\bar{q}] = 1 \mathbf{do } S, \rho \rangle \rightarrow \langle E, M_0 \rho M_0^\dagger \rangle}$$

$$(Loop 1) \quad \overline{\langle \mathbf{while } M[\bar{q}] = 1 \mathbf{do } S, \rho \rangle \rightarrow \langle S; \mathbf{while } M[\bar{q}] = 1 \mathbf{do } S, M_1 \rho M_1^\dagger \rangle}$$

Fig. 1. Transitional semantics of quantum programs.

the q system, as follows:

$$(|\varphi\rangle_q \langle \psi|) |\varphi'\rangle_q = \langle \psi | \varphi' \rangle |\varphi\rangle_q$$

for every state $|\varphi'\rangle$ of the q system. Sometimes, $|\varphi\rangle_q \langle \psi|$ should be understood as its cylinder extension on $\mathcal{H}_{all}$ according to Eq. (1). Similarly, we can define $|\varphi\rangle_{\bar{q}}$ and $|\varphi\rangle_{\bar{q}} \langle \psi|$ when $\bar{q}$ is a quantum register and $|\varphi\rangle$ and $|\psi\rangle$ are states in $\mathcal{H}_{\bar{q}}$. To see the role of initialization more clearly, we consider the case of $type(q) = \mathbf{integer}$ as an example. First, we examine a special case where ρ is a pure state; that is, $\rho = |\psi\rangle \langle \psi|$ for some $|\psi\rangle \in \mathcal{H}_{all}$. We can write $|\psi\rangle$ in the form:

$$|\psi\rangle = \sum_k \alpha_k |\psi_k\rangle,$$

where $|\psi_k\rangle$ is a product state for every k , say

$$|\psi_k\rangle = \bigotimes_{\text{all } q'} |\psi_{kq'}\rangle.$$

Then

$$\rho = \sum_{k,l} \alpha_k \alpha_l^* |\psi_k\rangle \langle \psi_l|.$$

After the initialization the state becomes:

$$\begin{aligned} \rho_0^q &= \sum_{n=-\infty}^{\infty} |0\rangle_q \langle n| \rho |n\rangle_q \langle 0| \\ &= \sum_{k,l} \alpha_k \alpha_l^* \left(\sum_{n=-\infty}^{\infty} |0\rangle_q \langle n| \psi_k \rangle \langle \psi_l | n \rangle_q \langle 0| \right) \\ &= \sum_{k,l} \alpha_k \alpha_l^* \left(\sum_{n=-\infty}^{\infty} \langle \psi_{lq} | n \rangle \langle n | \psi_{kq} \rangle \right) \left(|0\rangle_q \langle 0| \otimes \bigotimes_{q' \neq q} |\psi_{kq'}\rangle \langle \psi_{lq'}| \right) \\ &= \sum_{k,l} \alpha_k \alpha_l^* \langle \psi_{lq} | \psi_{kq} \rangle \left(|0\rangle_q \langle 0| \otimes \bigotimes_{q' \neq q} |\psi_{kq'}\rangle \langle \psi_{lq'}| \right) \\ &= |0\rangle_q \langle 0| \otimes \left(\sum_{k,l} \alpha_k \alpha_l^* \langle \psi_{lq} | \psi_{kq} \rangle \bigotimes_{q' \neq q} |\psi_{kq'}\rangle \langle \psi_{lq'}| \right). \end{aligned} \quad (2)$$

In general, suppose ρ is generated by an ensemble $\{(p_i, |\psi_i\rangle)\}$ of pure states; that is,

$$\rho = \sum_i p_i |\psi_i\rangle \langle \psi_i|.$$

For each i , we write $\rho_i = |\psi_i\rangle \langle \psi_i|$ and assume that it becomes ρ_{i0}^q after the initialization. By the above argument, we can write ρ_{i0} in the following form:

$$\rho_{i0}^q = \sum_k \alpha_{ik} (|0\rangle_q \langle 0| \otimes |\varphi_{ik}\rangle \langle \varphi_{ik}|),$$

where $|\varphi_{ik}\rangle \in \mathcal{H}_{Var \setminus \{q\}}$ for all k . Then the initialization makes that ρ become

$$\begin{aligned} \rho_0^q &= \sum_{n=-\infty}^{\infty} |0\rangle_q \langle n| \rho |n\rangle_q \langle 0| \\ &= \sum_i p_i \left(\sum_{n=-\infty}^{\infty} |0\rangle_q \langle n| \rho_i |n\rangle_q \langle 0| \right) \\ &= \sum_{i,k} p_i \alpha_{ik} (|0\rangle_q \langle 0| \otimes |\varphi_{ik}\rangle \langle \varphi_{ik}|). \end{aligned} \quad (3)$$

From Eqs. (2) and (3) we see that the state of variable q is set to be $|0\rangle$ and the states of the other quantum variables are unchanged. The statement “ $\bar{q} := U\bar{q}$ ” simply means that unitary transformation U is performed on quantum register $\bar{q}$, leaving the states of the quantum variables not in $\bar{q}$ unchanged. Note that U in the target configuration of the rule (Unitary Transformation) indeed stands for the cylinder extension of U

on $\mathcal{H}_{all}$ (see Eq. (1)). A similar remark applies to the rules for measurements and loops. Sequential composition is similar to its counterpart in classical computation. The program construct “**measure** $M[\bar{q}] : \bar{S}$ ” is a quantum generalization of a classical conditional statement. Recall that the first step in the execution of the conditional statement “**if** B **then** S_1 **else** S_2 ” is to check whether Boolean expression B is satisfied. However, according to a basic postulate of quantum mechanics, the unique way to acquire information about a quantum system is to measure it. So, in executing the statement “**measure** $M[\bar{q}] : \bar{S}$ ”, quantum measurement M will first be performed on quantum register $\bar{q}$, and then a subprogram S_m in $\bar{S}$ will be selected to be executed next according to the outcome of the measurement. The essential difference between a measurement statement and a classical conditional statement is that the state of program variables is changed after performing the measurement in the former, whereas it is not changed after checking the Boolean expression in the latter. Note that the outcome m is observed with probability

$$p_m = \text{tr}(M_m \rho M_m^\dagger),$$

and after the measurement the state becomes

$$\rho_m = M_m \rho M_m^\dagger / p_m.$$

So a natural presentation of the Measurement rule is the following probabilistic transition:

$$\langle \text{measure } M[\bar{q}] : \bar{S}, \rho \rangle \xrightarrow{p_m} \langle S_m, \rho_m \rangle$$

However, we adopt Selinger’s suggestion [Selinger 2004b] of encoding both probability p_m and density operator ρ_m into the partial density operator

$$M_m \rho M_m^\dagger = p_m \rho_m.$$

This allows us to give the measurement rule in terms of ordinary (nonprobabilistic) transition. It is worth noting that the probability p_m can be retrieved by taking the trace of the partial density operator in the target configuration of this rule. The statement “**while** $M[\bar{q}] = 1$ **do** S ” is a quantum generalization of classical loop “**while** B **do** S ”. To acquire information about quantum register $\bar{q}$, a measurement M is performed on it. The measurement M is a yes-no measurement with only two possible outcomes 0, 1. If the outcome 0 (no) is observed, then the program terminates, and if the outcome 1 (yes) occurs, then the program executes the subprogram S and continues. The only difference between a quantum loop and a classical loop is that checking the loop guard B in a classical loop does not change the state of program variables, but in a quantum loop the measurement outcomes 0 and 1 occur with the probabilities:

$$p_0 = \text{tr}(M_0 \rho M_0^\dagger), \quad p_1 = \text{tr}(M_1 \rho M_1^\dagger),$$

respectively, and the state becomes $M_0 \rho M_0^\dagger$ from ρ when the outcome is 0, and it becomes $M_1 \rho M_1^\dagger$ when the outcome is 1. Again, we adopt Selinger’s suggestion so that the (Loop 0) and (Loop 1) rules can be stated as ordinary transitions instead of probabilistic transitions.

Let S be a quantum program and $\rho \in \mathcal{D}^-(\mathcal{H})$. If $\langle S', \rho' \rangle$ can be reached from $\langle S, \rho \rangle$ in n steps in the transition relation $\rightarrow$; that is, there are configurations $\langle S_1, \rho_1 \rangle, \dots, \langle S_{n-1}, \rho_{n-1} \rangle$ such that

$$\langle S, \rho \rangle \rightarrow \langle S_1, \rho_1 \rangle \rightarrow \dots \rightarrow \langle S_{n-1}, \rho_{n-1} \rangle \rightarrow \langle S', \rho' \rangle,$$

then we write

$$\langle S, \rho \rangle \rightarrow^n \langle S', \rho' \rangle.$$

A transition sequence of S starting in ρ is a finite or infinite sequence of configurations in the following form:

$$\langle S, \rho \rangle \rightarrow \langle S_1, \rho_1 \rangle \rightarrow \cdots \rightarrow \langle S_n, \rho_n \rangle \rightarrow \langle S_{n+1}, \rho_{n+1} \rangle \rightarrow \cdots.$$

If it cannot be extended, then it is called a computation of S starting in ρ . Moreover, if it is finite and its last configuration is $\langle E, \rho' \rangle$, then we say that it terminates in ρ' ; and if it is infinite, then we say that it diverges. We say that S can diverge from ρ whenever it has a diverging computation starting in ρ .

Classical **while**-programs are a typical class of deterministic programs that have exactly one computation starting in a given state. As shown in the following example, however, quantum **while**-programs no longer possess such a determinism because probabilism is introduced by the measurements in the statements “**measure** $M[\bar{q}] : \bar{S}$ ” and “**while** $M[\bar{q}] = 1$ **do** S ”. After encoding probabilities into partial density operators, probabilism manifests as nondeterminism in transition rules (Measurement), (Loop 0) and (Loop 1).

Example 4.1. Suppose that $\text{type}(q_1) = \mathbf{Boolean}$ and $\text{type}(q_2) = \mathbf{integer}$. Consider the program:

$$S \equiv q_1 := 0; q_2 := 0; q_1 := Hq_1; q_2 := q_2 + 2; \mathbf{measure} \ M[q_1] : \bar{S}$$

where:

- M is the measurement according to the computational basis $\{|0\rangle, |1\rangle\}$ of $\mathcal{H}_2$; that is, $M = \{M_0, M_1\}$, $M_0 = |0\rangle\langle 0|$ and $M_1 = |1\rangle\langle 1|$;
- $\bar{S} \equiv S_1, S_2$, and
 - $S_1 \equiv \mathbf{skip}$;
 - $S_2 \equiv \mathbf{while} \ N[q_2] = 1 \ \mathbf{do} \ q_1 := \sigma_z q_1$, where σ_z is a Pauli matrix (see Example 2.2 (1)), $N = \{N_0, N_1\}$,

$$N_0 = \sum_{n=-\infty}^0 |n\rangle\langle n| \text{ and } N_1 = \sum_{n=1}^{\infty} |n\rangle\langle n|.$$

Let

$$\rho_0 = \bigotimes_{q \neq q_1, q_2} |0\rangle_q \langle 0|$$

and

$$\rho = |1\rangle_{q_1} \langle 1| \otimes | -1\rangle_{q_2} \langle -1| \otimes \rho_0.$$

Then the computations of S starting in ρ are

$$\begin{aligned} \langle S, \rho \rangle &\rightarrow \langle q_2 := 0; q_1 := Hq_1; q_2 := q_2 + 2; \mathbf{measure}, \rho_1 \rangle \\ &\rightarrow \langle q_1 := Hq_1; q_2 := q_2 + 2; \mathbf{measure}, \rho_2 \rangle \\ &\rightarrow \langle q_2 := q_2 + 2; \mathbf{measure}, \rho_3 \rangle \\ &\rightarrow \langle \mathbf{measure}, \rho_4 \rangle \\ &\rightarrow \begin{cases} \langle S_1, \rho_5 \rangle \rightarrow \langle E, \rho_5 \rangle, \\ \langle S_2, \rho_6 \rangle, \end{cases} \end{aligned}$$

$$\begin{aligned}
\langle S_2, \rho_6 \rangle &\rightarrow \langle q_1 := \sigma_z q_1; S_2, \rho_6 \rangle \\
&\rightarrow \langle S_2, -\rho_6 \rangle \\
&\rightarrow \dots \\
&\rightarrow^{2n-1} \langle q_1 := \sigma_z q_1; S_2, (-1)^{n-1} \rho_6 \rangle \\
&\rightarrow \langle S_2, (-1)^n \rho_6 \rangle \\
&\rightarrow \dots
\end{aligned}$$

where **measure** stands for the statement “**measure** $M[q_1] : \bar{S}$ ”, and

$$\begin{aligned}
\rho_1 &= |0\rangle_{q_1} \langle 0| \otimes |-1\rangle_{q_2} \langle -1| \otimes \rho_0, \\
\rho_2 &= |0\rangle_{q_1} \langle 0| \otimes |0\rangle_{q_2} \langle 0| \otimes \rho_0, \\
\rho_3 &= |+\rangle_{q_1} \langle +| \otimes |0\rangle_{q_2} \langle 0| \otimes \rho_0, \\
\rho_4 &= |+\rangle_{q_1} \langle +| \otimes |2\rangle_{q_2} \langle 2| \otimes \rho_0, \\
\rho_5 &= \frac{1}{2} |0\rangle_{q_1} \langle 0| \otimes |2\rangle_{q_2} \langle 2| \otimes \rho_0, \\
\rho_6 &= \frac{1}{2} |1\rangle_{q_1} \langle 1| \otimes |2\rangle_{q_2} \langle 2| \otimes \rho_0.
\end{aligned}$$

So S can diverge from ρ . Note that S_2 also has the transition

$$\langle S_2, (-1)^n \rho_6 \rangle \rightarrow \langle E, 0_{\mathcal{H}_{all}} \rangle,$$

but we always discard the transitions in which the partial density operator of the target configuration is the zero operator.

5. DENOTATIONAL SEMANTICS OF QUANTUM PROGRAMS

The denotational semantics of a quantum program is defined as a semantic function which maps partial density operators to themselves. More precisely, for any quantum program S , the semantic function of S sums the computed results of all terminating computations of S .

We write $\rightarrow^*$ for the reflexive and transitive closures of $\rightarrow$; that is, $\langle S, \rho \rangle \rightarrow^* \langle S', \rho' \rangle$ if and only if $\langle S, \rho \rangle \rightarrow^n \langle S', \rho' \rangle$ for some $n \geq 0$.

Definition 5.1. Let S be a quantum program. Then its semantic function

$$\llbracket S \rrbracket : \mathcal{D}^-(\mathcal{H}_{all}) \rightarrow \mathcal{D}^-(\mathcal{H}_{all})$$

is defined by

$$\llbracket S \rrbracket(\rho) = \sum \{ |\rho' : \langle S, \rho \rangle \rightarrow^* \langle E, \rho' \rangle| \} \quad (4)$$

for all $\rho \in \mathcal{D}^-(\mathcal{H}_{all})$.

It should be pointed out that $\{|\cdot|\}$ in Eq. (4) stands for a multiset. The reason for using a multiset is that the same density operator may be obtained through different computational paths, as we can see from the measurement and loop rules in the operational semantics. The following simple example illustrates the case more explicitly.

Example 5.1. Assume that $\text{type}(q) = \mathbf{Boolean}$. Consider the program:

$$S \equiv q := 0; q := Hq; \mathbf{measure} \ M[q] : \bar{S},$$

where:

- M is the measurement according to the computational basis $\{|0\rangle, |1\rangle\}$ of $\mathcal{H}_2$;
- $\bar{S} \equiv S_0, S_1$, in which $S_0 \equiv q := Iq$, I is the identity operator on qubits, $S_1 \equiv q := \sigma_x q$, and σ_x is a Pauli matrix (see Example 2.2 (1)).

Let $\rho = |0\rangle_{\text{all}}\langle 0|$, where

$$|0\rangle_{\text{all}} = \bigotimes_{\text{all } q} |0\rangle_q.$$

Then the computation of S starting in ρ is given as follows:

$$\begin{aligned} \langle S, \rho \rangle &\rightarrow \langle q := Hq; \mathbf{measure}, \rho \rangle \\ &\rightarrow \langle \mathbf{measure}, |+\rangle_q \left\langle +| \otimes \bigotimes_{q' \neq q} |0\rangle_{q'} \right\rangle \langle 0| \rangle \\ &\rightarrow \begin{cases} \langle S_0, \frac{1}{2}|0\rangle_q \langle 0| \otimes \bigotimes_{q' \neq q} |0\rangle_{q'} \langle 0| \rangle \rightarrow \langle E, \frac{1}{2}\rho \rangle, \\ \langle S_1, \frac{1}{2}|1\rangle_q \langle 1| \otimes \bigotimes_{q' \neq q} |0\rangle_{q'} \langle 0| \rangle \rightarrow \langle E, \frac{1}{2}\rho \rangle, \end{cases} \end{aligned}$$

where **measure** is an abbreviation of statement “**measure** $M[q] : \bar{S}$ ”. So we have

$$\llbracket S \rrbracket(\rho) = \frac{1}{2}\rho + \frac{1}{2}\rho = \rho.$$

Now we will establish some basic properties of semantic functions. First, we prove their linearity.

LEMMA 5.1. *Let $\rho_1, \rho_2 \in \mathcal{D}^-(D)$ and $\lambda_1, \lambda_2 \geq 0$. If $\lambda_1\rho_1 + \lambda_2\rho_2 \in \mathcal{D}^-(D)$, then for any quantum program S , we have*

$$\llbracket S \rrbracket(\lambda_1\rho_1 + \lambda_2\rho_2) = \lambda_1\llbracket S \rrbracket(\rho_1) + \lambda_2\llbracket S \rrbracket(\rho_2).$$

PROOF. We can easily prove the following fact by induction on the structure of S :

—*Claim:* If $\langle S, \rho_1 \rangle \rightarrow \langle S', \rho'_1 \rangle$ and $\langle S, \rho_2 \rangle \rightarrow \langle S', \rho'_2 \rangle$, then

$$\langle S, \lambda_1\rho_1 + \lambda_2\rho_2 \rangle \rightarrow \langle S', \lambda_1\rho'_1 + \lambda_2\rho'_2 \rangle.$$

Then the conclusion follows immediately. $\square$

Next, we give a representation of the semantic function $\llbracket S \rrbracket$ according to the structure of the program S . To do this for quantum loops, we need some auxiliary notation. Let Ω be a quantum program such that $\llbracket \Omega \rrbracket = 0_{\mathcal{H}_{\text{all}}}$ for all $\rho \in \mathcal{D}(\mathcal{H})$; for example,

$$\Omega \equiv \mathbf{while } M_{\text{trivial}}[q] = 1 \mathbf{ do skip},$$

where q is a quantum variable, and

$$M_{\text{trivial}} = \{M_0 = 0_{\mathcal{H}_q}, M_1 = I_{\mathcal{H}_q}\}$$

is a trivial measurement on $\mathcal{H}_q$. We set

$$(\mathbf{while } M[\bar{q}] = 1 \mathbf{ do } S)^0 \equiv \Omega,$$

$$(\mathbf{while } M[\bar{q}] = 1 \mathbf{ do } S)^{n+1} \equiv \mathbf{measure } M[\bar{q}] : \bar{S},$$

where $\bar{S} \equiv S_0, S_1$, and

$$S_0 \equiv \mathbf{skip},$$

$$S_1 \equiv S; (\mathbf{while } M[\bar{q}] = 1 \mathbf{ do } S)^n$$

for all $n \geq 0$.

PROPOSITION 5.1.

(1) $\llbracket \mathbf{skip} \rrbracket(\rho) = \rho$.

(2) If $\text{type}(q) = \mathbf{Boolean}$, then

$$\llbracket q := 0 \rrbracket(\rho) = |0\rangle_q \langle 0| \rho |0\rangle_q \langle 0| + |0\rangle_q \langle 1| \rho |1\rangle_q \langle 0|;$$

and if $\text{type}(q) = \mathbf{integer}$, then

$$\llbracket q := 0 \rrbracket(\rho) = \sum_{n=-\infty}^{\infty} |0\rangle_q \langle n| \rho |n\rangle_q \langle 0|.$$

(3) $\llbracket \bar{q} := U\bar{q} \rrbracket(\rho) = U\rho U^\dagger$.

(4) $\llbracket S_1; S_2 \rrbracket(\rho) = \llbracket S_2 \rrbracket(\llbracket S_1 \rrbracket(\rho))$.

(5) $\llbracket \mathbf{measure} M[\bar{q}] : \bar{S} \rrbracket(\rho) = \sum_m \llbracket S_m \rrbracket(M_m \rho M_m^\dagger)$.

(6) $\llbracket \mathbf{while} M[\bar{q}] = 1 \mathbf{do} S \rrbracket(\rho) = \bigvee_{n=0}^{\infty} \llbracket (\mathbf{while} M[\bar{q}] = 1 \mathbf{do} S)^n \rrbracket(\rho)$, where $\bigvee$ stands for the least upper bound of partial density operators according to the Löwner partial order $\sqsubseteq$.

PROOF. (1), (2), and (3) are obvious.

(4) By Lemma 5.1 and the transition rule for sequential composition, we obtain

$$\begin{aligned} \llbracket S_2 \rrbracket(\llbracket S_1 \rrbracket(\rho)) &= \llbracket S_2 \rrbracket \left(\sum \{ |\rho_1 : \langle S_1, \rho \rangle \rightarrow^* \langle E, \rho_1 \rangle| \} \right) \\ &= \sum \{ \llbracket S_2 \rrbracket(\rho_1) : \langle S_1, \rho \rangle \rightarrow^* \langle E, \rho_1 \rangle \} \\ &= \sum \left\{ \left| \sum \{ |\rho' : \langle S_2, \rho_1 \rangle \rightarrow^* \langle E, \rho' \rangle| \} : \langle S_1, \rho \rangle \rightarrow^* \langle E, \rho_1 \rangle \right| \right\} \\ &= \sum \{ |\rho' : \langle S_1, \rho \rangle \rightarrow^* \langle E, \rho_1 \rangle \text{ and } \langle S_2, \rho_1 \rangle \rightarrow^* \langle E, \rho' \rangle| \} \\ &= \sum \{ |\rho' : \langle S_1; S_2, \rho \rangle \rightarrow^* \langle E, \rho' \rangle| \} \\ &= \llbracket S_1; S_2 \rrbracket(\rho). \end{aligned}$$

(5) follows immediately from the transition rule for measurement.

(6) We introduce two auxiliary operators:

$$\mathcal{E}_i : \mathcal{D}^-(\mathcal{H}_{all}) \rightarrow \mathcal{D}^-(\mathcal{H}_{all}),$$

$$\mathcal{E}_i(\rho) = M_i \rho M_i^\dagger$$

for all $\rho \in \mathcal{D}^-(\mathcal{H})$ and $i = 0, 1$. For simplicity, we write **while** for statement “**while** $M[\bar{q}] = 1$ **do** S ”. First, we prove that

$$\llbracket (\mathbf{while})^n \rrbracket(\rho) = \sum_{k=0}^{n-1} [\mathcal{E}_0 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^k](\rho)$$

for all $n \geq 1$ by induction on n . The case of $n = 1$ is obvious. Then, by (1), (4), and (5) proved above and the induction hypothesis on $n - 1$, we obtain

$$\begin{aligned} \llbracket (\mathbf{while})^n \rrbracket(\rho) &= \llbracket \mathbf{skip} \rrbracket(\mathcal{E}_0(\rho)) + \llbracket S; (\mathbf{while})^n \rrbracket(\mathcal{E}_1(\rho)) \\ &= \mathcal{E}_0(\rho) + \llbracket (\mathbf{while})^{n-1} \rrbracket(\llbracket S \rrbracket \circ \mathcal{E}_1(\rho)) \\ &= \mathcal{E}_0(\rho) + \sum_{k=0}^{n-2} [\mathcal{E}_0 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^k](\llbracket S \rrbracket \circ \mathcal{E}_0(\rho)) \\ &= \sum_{k=0}^{n-1} [\mathcal{E}_0 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^k](\rho). \end{aligned} \tag{5}$$

Second, it follows immediately from Eq. (5) that $\{\llbracket(\mathbf{while})^n\rrbracket(\rho)\}_{n=0}^\infty$ is an increasing sequence. Selinger proved that the set of partial density operators in a finite-dimensional Hilbert space equipped with the Löwner partial order $\sqsubseteq$ is a complete partial order (CPO), that is, a partially ordered set in which the least upper bound of every increasing sequence exists (see Selinger [2004b]; Proposition 3.6). This conclusion was generalized by the author to the case of infinite-dimensional Hilbert space (see Ying et al. [2010]; Proposition 2.1). Therefore,

$$\bigvee_{n=0}^\infty \llbracket(\mathbf{while})^n\rrbracket(\rho)$$

exists. Furthermore, we have

$$\begin{aligned} \llbracket\mathbf{while}\rrbracket(\rho) &= \sum \{|\rho' : \langle \mathbf{while}, \rho \rangle \rightarrow^* \langle E, \rho' \rangle|\} \\ &= \sum_{n=1}^\infty \sum \{|\rho' : \langle \mathbf{while}, \rho \rangle \rightarrow^n \langle E, \rho' \rangle|\}. \end{aligned}$$

So it suffices to show that

$$\sum \{|\rho' : \langle \mathbf{while}, \rho \rangle \rightarrow^n \langle E, \rho' \rangle|\} = [\mathcal{E}_0 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^{n-1}](\rho)$$

for all $n \geq 1$. This can be easily done by induction on n . $\square$

The denotational semantics of quantum programs has been considered in the previous literature (e.g., Selinger [2004b] and Feng et al. [2007]). If we only consider quantum variables of type **Boolean**, then the above proposition coincides with Figure 1 in Feng et al. [2007]. However, it is worth noting that in Feng et al. [2007] the denotational semantics of quantum programs was defined directly, and it lacks a basis of operational semantics. Indeed, the operational/denotation correspondence given in Definition 5.1 and the above proposition also extends the results presented in Selinger [2004b] where no operational semantics was defined. Similar to Lemma 3.2 in Feng et al. [2007], we may prove that the semantic function of a quantum program is a super-operator (see Nielsen and Chuang [2000]; Section 8.2). Thus, the denotational semantics given in this section is consistent with Selinger's idea [Selinger 2004b] of modeling quantum programs as super-operators.

A recursive characterization of the semantic function of a quantum loop can be derived from the preceding proposition.

COROLLARY 5.1. *If we write **while** for quantum loop “**while** $M[\bar{q}] = 1$ **do** S ”, then for any $\rho \in \mathcal{D}^-(\mathcal{H}_{all})$, it holds that*

$$\llbracket\mathbf{while}\rrbracket(\rho) = M_0 \rho M_0^\dagger + \llbracket\mathbf{while}\rrbracket(\llbracket S \rrbracket(M_1 \rho M_1^\dagger)).$$

PROOF. Immediate from Proposition 5.1, (6) and Eq. (5). $\square$

The following proposition shows that a semantic function does not increase the trace of density operator of quantum variables.

PROPOSITION 5.2. *For any quantum program S , it holds that*

$$\text{tr}(\llbracket S \rrbracket(\rho)) \leq \text{tr}(\rho)$$

for all $\rho \in \mathcal{D}^-(\mathcal{H}_{all})$.

PROOF. We proceed by induction on the structure of S .

Case 1. $S \equiv \mathbf{skip}$. Obvious.

Case 2. $S \equiv q := 0$. If $\text{type}(q) = \mathbf{integer}$, then

$$\begin{aligned} \text{tr}(\llbracket S \rrbracket(\rho)) &= \sum_{n=-\infty}^{\infty} \text{tr}(|0\rangle_q \langle n| \rho |n\rangle_q \langle 0|) \\ &= \sum_{n=-\infty}^{\infty} \text{tr}_q(|0\rangle_q \langle n| \rho |n\rangle_q) \\ &= \text{tr} \left[\left(\sum_{n=-\infty}^{\infty} |n\rangle_q \langle n| \right) \rho \right] \\ &= \text{tr}(\rho). \end{aligned}$$

It can be proved in a similar way when $\text{type}(q) = \mathbf{Boolean}$.

Case 3. $S \equiv \bar{q} := U\bar{q}$. Then

$$\text{tr}(\llbracket S \rrbracket(\rho)) = \text{tr}(U\rho U^\dagger) = \text{tr}(U^\dagger U\rho) = \text{tr}(\rho).$$

Case 4. $S \equiv S_1; S_2$. It follows from the induction hypothesis on S_1 and S_2 that

$$\begin{aligned} \text{tr}(\llbracket S \rrbracket(\rho)) &= \text{tr}(\llbracket S_2 \rrbracket(\llbracket S_1 \rrbracket(\rho))) \\ &\leq \text{tr}(\llbracket S_1 \rrbracket(\rho)) \\ &\leq \text{tr}(\rho). \end{aligned}$$

Case 5. $S \equiv \mathbf{measure} \ M[\bar{q}] : \bar{S}$. Then, by induction hypothesis, we obtain

$$\begin{aligned} \text{tr}(\llbracket S \rrbracket(\rho)) &= \sum_m \text{tr}(\llbracket S_m \rrbracket(M_m \rho M_m^\dagger)) \\ &\leq \sum_m \text{tr}(M_m \rho M_m^\dagger) \\ &= \text{tr} \left[\left(\sum_m M_m^\dagger M_m \right) \rho \right] \\ &= \text{tr}(\rho). \end{aligned}$$

Case 6. $S \equiv \mathbf{while} \ M[\bar{q}] = 1 \ \mathbf{do} \ S'$. We write $(\mathbf{while})^n$ for statement “ $(\mathbf{while} \ M[\bar{q}] = 1 \ \mathbf{do} \ S')^n$ ”. With Proposition 5.1, (6), it suffices to show that

$$\text{tr}(\llbracket (\mathbf{while})^n \rrbracket(\rho)) \leq \text{tr}(\rho)$$

for all $n \geq 0$. This can be carried out by induction on n . The case of $n = 0$ is obvious. By the induction hypothesis on n and S' , we have

$$\begin{aligned} \text{tr}(\llbracket (\mathbf{while})^{n+1} \rrbracket(\rho)) &= \text{tr}(M_0 \rho M_0^\dagger) + \text{tr}(\llbracket (\mathbf{while})^n \rrbracket(\llbracket S' \rrbracket(M_1 \rho M_1^\dagger))) \\ &\leq \text{tr}(M_0 \rho M_0^\dagger) + \text{tr}(\llbracket S' \rrbracket(M_1 \rho M_1^\dagger)) \\ &\leq \text{tr}(M_0 \rho M_0^\dagger) + \text{tr}(M_1 \rho M_1^\dagger) \\ &= \text{tr}[(M_0^\dagger M_0 + M_1^\dagger M_1) \rho] \\ &= \text{tr}(\rho). \end{aligned}$$

□

From the proof of the preceding proposition it is easy to see that $\text{tr}(\llbracket S \rrbracket(\rho)) = \text{tr}(\rho)$ whenever S does not contain any quantum loop. The unique possibility that $\text{tr}(\llbracket S \rrbracket(\rho)) < \text{tr}(\rho)$ comes from the quantum loops occurring in S . Thus, $\text{tr}(\rho) - \text{tr}(\llbracket S \rrbracket(\rho))$ is the

probability that program S diverges from input state ρ . This can be further illustrated by the following example.

Example 5.2. Let $\text{type}(q) = \mathbf{integer}$, and let

$$M_0 = \sum_{n=1}^{\infty} \sqrt{\frac{n-1}{2n}} (|n\rangle\langle n| + |-n\rangle\langle -n|),$$

$$M_1 = \sum_{n=1}^{\infty} \sqrt{\frac{n+1}{2n}} (|n\rangle\langle n| + |-n\rangle\langle -n|) + |0\rangle\langle 0|.$$

Then $M = \{M_0, M_1\}$ is a yes-no measurement on the state space $\mathcal{H}_q$ of quantum variable q . Consider the quantum loop:

while $M[q] = 1$ **do** $q := q + 1$.

For simplicity, we write **while** for this program. Let

$$\rho_0 = \bigotimes_{q' \neq q} |0\rangle_{q'} \langle 0|$$

and $\rho = |0\rangle_q \langle 0| \otimes \rho_0$. Then

$$\llbracket (\mathbf{while})^n \rrbracket(\rho) = \begin{cases} 0_{\mathcal{H}_{all}} & \text{if } n = 0, 1, 2, \\ \frac{1}{2} \left(\sum_{k=2}^{n-1} \frac{k-1}{k!} |k\rangle_q \langle k| \right) \otimes \rho_0 & \text{if } n \geq 3, \end{cases}$$

$$\llbracket \mathbf{while} \rrbracket(\rho) = \frac{1}{2} \left(\sum_{n=2}^{\infty} \frac{n-1}{n!} |n\rangle_q \langle n| \right) \otimes \rho_0,$$

and

$$\text{tr}(\llbracket \mathbf{while} \rrbracket(\rho)) = \frac{1}{2} \sum_{n=2}^{\infty} \frac{n-1}{n!} = \frac{1}{2}.$$

This means that program **while** terminates on input ρ with probability $\frac{1}{2}$, and it diverges from input ρ with probability $\frac{1}{2}$.

To conclude this section, we observe how quantum programs change the states of quantum variables and how they access quantum variables.

Let $X \subseteq \text{Var}$ be a set of quantum variables. For any operator $A \in \mathcal{L}(\mathcal{H}_{all})$, we write $\text{tr}_X(A)$ for the partial trace

$$\text{tr}_{\bigotimes_{q \in X} \mathcal{H}_q}(A).$$

PROPOSITION 5.3.

- (1) $\text{tr}_{\text{var}(S)}(\llbracket S \rrbracket(\rho)) = \text{tr}_{\text{var}(S)}(\rho)$.
- (2) If $\text{tr}_{\text{Var} \setminus \text{var}(S)}(\rho_1) = \text{tr}_{\text{Var} \setminus \text{var}(S)}(\rho_2)$, then

$$\text{tr}_{\text{Var} \setminus \text{var}(S)}(\llbracket S \rrbracket(\rho_1)) = \text{tr}_{\text{Var} \setminus \text{var}(S)}(\llbracket S \rrbracket(\rho_2)).$$

We put the long and tedious proof of the above proposition in the Appendix.

Recall that $\text{tr}_X(\rho)$ describes the state of the quantum variables not in X when the global state of all quantum variables is ρ . So Proposition 5.3, (1) indicates that the state of the quantum variables not in $\text{var}(S)$, after implementing quantum program S , is the same as that before implementing S . This means that program S can

only change the state of quantum variables in $\text{var}(S)$. On the other hand, Proposition 5.3, (2) shows that if two input states ρ_1 and ρ_2 coincide on the quantum variables in $\text{var}(S)$, then the computed outcomes of S , starting in ρ_1 and ρ_2 , respectively, will also coincide on these quantum variables. In other words, program S can access at most the quantum variables in $\text{var}(S)$.

6. CORRECTNESS FORMULAS

The correctness of a quantum program will be expressed by a quantum extension of the Hoare triple, in which a quantum predicate describes the input state and a quantum predicate describes the output states of the program. We adopt D'Hondt and Panangaden's definition of quantum predicates [D'Hondt and Panangaden 2006]. For any $X \subseteq \text{Var}$, a quantum predicate on $\mathcal{H}_X$ is defined to be a Hermitian operator P on $\mathcal{H}_X$ such that

$$0_{\mathcal{H}_X} \sqsubseteq P \sqsubseteq I_{\mathcal{H}_X}.$$

We write $\mathcal{P}(\mathcal{H}_X)$ for the set of quantum predicates on $\mathcal{H}_X$. Intuitively, for any $\rho \in \mathcal{D}^-(\mathcal{H}_X)$, $\text{tr}(P\rho)$ stands for the probability that predicate P is satisfied in state ρ .

A correctness formula is a statement of the form

$$\{P\}S\{Q\}$$

where S is a quantum program, and both P and Q are quantum predicates on $\mathcal{H}_{\text{all}}$. The quantum predicate P is called the precondition of the correctness formula and Q the postcondition. A correctness formula can be interpreted in two different ways:

Definition 6.1.

- (1) The correctness formula $\{P\}S\{Q\}$ is true in the sense of total correctness, written

$$\models_{\text{tot}} \{P\}S\{Q\},$$

if we have

$$\text{tr}(P\rho) \leq \text{tr}(Q\llbracket S \rrbracket(\rho))$$

for all $\rho \in \mathcal{D}^-(\mathcal{H})$.

- (2) The correctness formula $\{P\}S\{Q\}$ is true in the sense of partial correctness, written

$$\models_{\text{par}} \{P\}S\{Q\},$$

if we have:

$$\text{tr}(P\rho) \leq \text{tr}(Q\llbracket S \rrbracket(\rho)) + [\text{tr}(\rho) - \text{tr}(\llbracket S \rrbracket(\rho))]$$

for all $\rho \in \mathcal{D}^-(\mathcal{H})$.

The intuitive meaning of the defining inequality of total correctness is: the probability that input ρ satisfies quantum predicate P is not greater than the probability that quantum program S terminates on ρ and its output $\llbracket S \rrbracket(\rho)$ satisfies quantum predicate Q . Recall that $\text{tr}(\rho) - \text{tr}(\llbracket S \rrbracket(\rho))$ is the probability that quantum program S diverges from input ρ . Thus, the definition inequality of partial correctness means that if input ρ satisfies predicate P , then either program S terminates on it and its output $\llbracket S \rrbracket(\rho)$ satisfies Q , or S diverges from it. The difference between total correctness and partial correctness is illustrated well by the following simple example.

Example 6.1. Assume that $\text{type}(q) = \mathbf{Boolean}$. Consider the following program:

$$S \equiv \mathbf{while} \ M[q] = 1 \ \mathbf{do} \ q := \sigma_z q$$

where M is the measurement in the computational basis $|0\rangle, |1\rangle$; that is, $M_0 = |0\rangle\langle 0|$ and $M_1 = |1\rangle\langle 1|$, and σ_z is a Pauli matrix. Let $P = |\psi\rangle_q \langle \psi| \otimes P'$, where $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \in \mathcal{H}_2$,

and $P' \in \mathcal{P}(\mathcal{H}_{\text{Var} \setminus \{q\}})$. Then

$$\models_{\text{tot}} \{P\}S\{|0\rangle_q\langle 0| \otimes P'\}$$

does not hold if $\beta \neq 0$ and $P \neq 0_{\mathcal{H}_{\text{Var} \setminus \{q\}}}$. In fact, put

$$\rho = |\psi\rangle_q\langle\psi| \otimes I_{\mathcal{H}_{\text{Var} \setminus \{q\}}}.$$

Then

$$\llbracket S \rrbracket(\rho) = |\alpha|^2 |0\rangle_q\langle 0| \otimes I_{\mathcal{H}_{\text{Var} \setminus \{q\}}}$$

and

$$\text{tr}(P\rho) = \text{tr}(P') > |\alpha|^2 \text{tr}(P') = \text{tr}(|0\rangle_q\langle 0| \otimes P') \llbracket S \rrbracket(\rho).$$

On the other hand, we always have

$$\models_{\text{par}} \{P\}S\{|0\rangle_q\langle 0| \otimes P'\}.$$

To show this, we first consider a special class of partial density operators on $\mathcal{H}_{\text{Var} \setminus \{q\}}$: $\rho = |\varphi\rangle_q\langle\varphi| \otimes \rho'$, where $|\varphi\rangle = a|0\rangle + b|1\rangle \in \mathcal{H}_2$, and $\rho' \in \mathcal{D}^-(\mathcal{H}_{\text{Var} \setminus \{q\}})$. A routine calculation yields

$$\llbracket S \rrbracket(\rho) = |a|^2 |0\rangle_q\langle 0| \otimes \rho'$$

and

$$\begin{aligned} \text{tr}(P\rho) &= |\langle\psi|\varphi\rangle|^2 \text{tr}(P'\rho') \\ &\leq |a|^2 \text{tr}(P'\rho') + [\text{tr}(\rho') - |a|^2 \text{tr}(\rho')] \\ &= \text{tr}(|0\rangle_q\langle 0| \otimes P') \llbracket S \rrbracket(\rho) + [\text{tr}(\rho) - \text{tr}(\llbracket S \rrbracket(\rho))]. \end{aligned}$$

Then it follows from linearity of $\llbracket S \rrbracket$ (see Lemma 5.1) that

$$\text{tr}(P\rho) \leq \text{tr}(|0\rangle_q\langle 0| \otimes P') \llbracket S \rrbracket(\rho) + [\text{tr}(\rho) - \text{tr}(\llbracket S \rrbracket(\rho))]$$

for all $\rho \in \mathcal{D}^-(\mathcal{H}_{\text{all}})$.

The following proposition presents some basic properties of correctness formulas.

PROPOSITION 6.1.

- (1) If $\models_{\text{tot}} \{P\}S\{Q\}$, then $\models_{\text{par}} \{P\}S\{Q\}$.
- (2) For any quantum program S , and for any $P, Q \in \mathcal{P}(\mathcal{H}_{\text{all}})$, we have
$$\models_{\text{tot}} \{0_{\mathcal{H}_{\text{all}}}\}S\{Q\}, \quad \models_{\text{par}} \{P\}S\{I_{\mathcal{H}_{\text{all}}}\}.$$
- (3) *Linearity:* For any $P_1, P_2, Q_1, Q_2 \in \mathcal{P}(\mathcal{H}_{\text{all}})$ and $\lambda_1, \lambda_2 \geq 0$ with $\lambda_1 P_1 + \lambda_2 P_2, \lambda_1 Q_1 + \lambda_2 Q_2 \in \mathcal{P}(\mathcal{H}_{\text{all}})$, if

$$\models_{\text{tot}} \{P_i\}S\{Q_i\} \quad (i = 1, 2),$$

then

$$\models_{\text{tot}} \{\lambda_1 P_1 + \lambda_2 P_2\}S\{\lambda_1 Q_1 + \lambda_2 Q_2\}.$$

The same conclusion holds for partial correctness if $\lambda_1 + \lambda_2 = 1$.

PROOF. Immediate from definition. $\square$

7. WEAKEST PRECONDITIONS AND WEAKEST LIBERAL PRECONDITIONS

As in the case of classical Floyd–Hoare logic, the notions of weakest precondition and weakest liberal precondition of a quantum program will play a key role in establishing the (relative) completeness of Floyd–Hoare logic for quantum programs. They may be defined in a familiar way:

Definition 7.1. Let S be a quantum program and $P \in \mathcal{P}(\mathcal{H}_{all})$ be a quantum predicate on $\mathcal{H}_{all}$.

- (1) The weakest precondition of S with respect to P is defined to be the quantum predicate $wp.S.P \in \mathcal{P}(\mathcal{H}_{all})$ satisfying the following conditions:
 - (a) $\models_{tot} \{wp.S.P\}S\{P\}$;
 - (b) if quantum predicate $Q \in \mathcal{P}(\mathcal{H}_{all})$ satisfies $\models_{tot} \{Q\}S\{P\}$, then $Q \sqsubseteq wp.S.P$.
- (2) The weakest liberal precondition of S with respect to P is defined to be the quantum predicate $wlp.S.P \in \mathcal{P}(\mathcal{H}_{all})$ satisfying the following conditions:
 - (a) $\models_{par} \{wlp.S.P\}S\{P\}$;
 - (b) if quantum predicate $Q \in \mathcal{P}(\mathcal{H}_{all})$ satisfies $\models_{par} \{Q\}S\{P\}$, then $Q \sqsubseteq wlp.S.P$.

The next two propositions give the explicit representations of weakest preconditions and weakest liberal preconditions, respectively. They will be used in the proof of completeness of the quantum Floyd–Hoare logic.

PROPOSITION 7.1.

- (1) (a) $wp.\mathbf{skip}.P = P$.
- (b) If $type(q) = \mathbf{Boolean}$, then

$$wp.(q := 0).P = |0\rangle_q \langle 0|P|0\rangle_q \langle 0| + |1\rangle_q \langle 0|P|0\rangle_q \langle 1|,$$

and if $type(q) = \mathbf{integer}$, then

$$wp.(q := 0).P = \sum_{n=-\infty}^{\infty} |n\rangle_q \langle 0|P|0\rangle_q \langle n|.$$

- (c) $wp.(\bar{q} := U\bar{q}).P = U^\dagger P U$.
- (d) $wp.(S_1; S_2).P = wp.S_1.(wp.S_2.P)$.
- (e) $wp.(\mathbf{measure} M[\bar{q}] : \bar{S}).P = \sum_m M_m^\dagger (wp.S_m.P) M_m$.
- (f) $wp.(\mathbf{while} M[\bar{q}] = 1 \mathbf{do} S).P = \bigvee_{n=0}^{\infty} P_n$, where

$$\begin{cases} P_0 = 0_{\mathcal{H}_{all}}, \\ P_{n+1} = M_0^\dagger P M_0 + M_1^\dagger (wp.S.P_n) M_1 \text{ for all } n \geq 0, \end{cases}$$

and $\bigvee$ stands for the least upper bound of quantum predicates according to the Löwner partial order $\sqsubseteq$.

- (2) For any quantum program S , for any quantum predicate $P \in \mathcal{P}(\mathcal{H}_{all})$, and for any partial density operator $\rho \in \mathcal{D}^-(\mathcal{H}_{all})$, we have

$$tr((wp.S.P)\rho) = tr(P\llbracket S \rrbracket(\rho)).$$

PROOF. The trick is to simultaneously prove (1) and (2) by induction on the structure of the program S . Indeed, this is why we put these two conclusions, which seem unconnected at the first glance into a single proposition.

Case 1. $S \equiv \mathbf{skip}$. Obvious.

Case 2. $S \equiv q := 0$. We only consider the case of $type(q) = \mathbf{integer}$, and the case of $type(q) = \mathbf{Boolean}$ is similar. First, it holds that

$$\begin{aligned} tr\left(\left(\sum_{n=-\infty}^{\infty} |n\rangle_q \langle 0|P|0\rangle_q \langle n|\right)\rho\right) &= tr\left(P \sum_{n=-\infty}^{\infty} |0\rangle_q \langle n|\rho|n\rangle_q \langle 0|\right) \\ &= tr(P\llbracket q := 0 \rrbracket(\rho)). \end{aligned}$$

On the other hand, for any quantum predicate $Q \in \mathcal{P}(\mathcal{H}_{all})$, if $\models_{tot} \{Q\}q := 0\{P\}$, that is,

$$\begin{aligned} tr(Q\rho) &\leq tr(P\llbracket q := 0 \rrbracket(\rho)) \\ &= tr\left(\left(\sum_{n=-\infty}^{\infty} |n\rangle_q \langle 0|P|0\rangle_q \langle n|\right)\rho\right) \end{aligned}$$

for all $\rho \in \mathcal{D}^-(\mathcal{H}_{all})$, then it follows from Lemma 2.1 that

$$Q \sqsubseteq \sum_{n=-\infty}^{\infty} |n\rangle_q \langle 0|P|0\rangle_q \langle n|.$$

Case 3. $S \equiv \bar{q} := U\bar{q}$. Similar to Case 2.

Case 4. $S \equiv S_1; S_2$. It follows from the induction hypothesis on S_1 and S_2 that

$$\begin{aligned} tr((wp.S_1.(wp.S_2.P))\rho) &= tr((wp.S_2.P)\llbracket S_1 \rrbracket(\rho)) \\ &= tr(P\llbracket S_2 \rrbracket(\llbracket S_1 \rrbracket(\rho))) \\ &= tr(P\llbracket S_1; S_2 \rrbracket(\rho)). \end{aligned}$$

If $\models_{tot} \{Q\}S_1; S_2\{P\}$, then for all $\rho \in \mathcal{D}^-(\mathcal{H}_{all})$, we have

$$tr(Q\rho) \leq tr(P\llbracket S_1; S_2 \rrbracket(\rho)) = tr((wp.S_1.(wp.S_2.P))\rho).$$

Therefore, it follows from Lemma 2.1 that $Q \sqsubseteq wp.S_1.(wp.S_2.P)$.

Case 5. $S \equiv \mathbf{measure} M[\bar{q}] : \bar{S}$. Applying the induction hypothesis on S_m , we obtain

$$\begin{aligned} tr\left(\left(\sum_m M_m^\dagger (wp.S_m.P) M_m\right)\rho\right) &= \sum_m tr((wp.S_m.P) M_m \rho M_m^\dagger) \\ &= \sum_m tr(P\llbracket S_m \rrbracket(M_m \rho M_m^\dagger)) \\ &= tr\left(P \sum_m \llbracket S_m \rrbracket(M_m \rho M_m^\dagger)\right) \\ &= tr(P\llbracket \mathbf{measure} M[\bar{q}] : \bar{S} \rrbracket_{\text{par}}(\rho)). \end{aligned}$$

If $\models_{tot} \{Q\}\mathbf{measure} M[\bar{q}] : \bar{S}\{P\}$, then

$$tr(Q\rho) \leq tr\left(\left(\sum_m M_m^\dagger (wp.S_m.P) M_m\right)\rho\right)$$

for all ρ , and it follows from Lemma 2.1 that

$$Q \sqsubseteq \sum_m M_m^\dagger (wp.S_m.P) M_m.$$

Case 6. $S \equiv \mathbf{while} M[\bar{q}] = 1 \mathbf{do} S'$. For simplicity, we write $(\mathbf{while})^n$ for statement “ $(\mathbf{while} M[\bar{q}] = 1 \mathbf{do} S')^n$ ”. First, we have, for any $\rho \in \mathcal{D}^-(\mathcal{H})$,

$$tr(P_n \rho) = tr(P\llbracket (\mathbf{while})^n \rrbracket(\rho)). \quad (6)$$

This claim can be proved by induction on n . The basis case of $n = 0$ is obvious. By the induction hypotheses on n and S' , we obtain

$$\begin{aligned}
tr(P_{n+1}\rho) &= tr(M_0^\dagger P M_0 \rho) + tr(M_1^\dagger (wp.S'.P_n) M_1 \rho) \\
&= tr(P M_0 \rho M_0^\dagger) + tr((wp.S'.P_n) M_1 \rho M_1^\dagger) \\
&= tr(P M_0 \rho M_0^\dagger) + tr(P_n \llbracket S' \rrbracket (M_1 \rho M_1^\dagger)) \\
&= tr(P M_0 \rho M_0^\dagger) + tr(P \llbracket (\mathbf{while})^n \rrbracket (\llbracket S' \rrbracket (M_1 \rho M_1^\dagger))) \\
&= tr[P(M_0 \rho M_0^\dagger + \llbracket S'; (\mathbf{while})^n \rrbracket (M_1 \rho M_1^\dagger))] \\
&= tr(P \llbracket (\mathbf{while})^{n+1} \rrbracket (\rho)).
\end{aligned}$$

Thus, Eq. (6) is proved. Second, we know from the proof of Proposition 5.1, (6) that

$$\{\llbracket (\mathbf{while})^n \rrbracket (\rho)\}_{n=0}^\infty$$

is an increasing sequence with respect to the Löwner partial order $\sqsubseteq$. On the other hand, P is a positive operator. So, for any $n \geq 0$, it holds that

$$tr(P_n \rho) = tr(P \llbracket (\mathbf{while})^n \rrbracket (\rho)) \leq tr(P \llbracket (\mathbf{while})^{n+1} \rrbracket (\rho)) = tr(P_{n+1} \rho). \quad (7)$$

Since ρ in Eq. (7) is arbitrary, we assert that $P_n \sqsubseteq P_{n+1}$ by Lemma 2.1, and $\{P_n\}_{n=0}^\infty$ is an increasing sequence with respect to the Löwner partial order. Furthermore, the set of quantum predicates equipped with the Löwner partial order is a complete partial order (CPO) (see D'Hondt and Panangaden [2006]; Proposition 2.3). Therefore, $\bigvee_{n=0}^\infty P_n$ exists. Now the continuity of a trace operator yields

$$\begin{aligned}
tr\left(\left(\bigvee_{n=0}^\infty P_n\right)\rho\right) &= \bigvee_{n=0}^\infty tr(P_n \rho) \\
&= \bigvee_{n=0}^\infty tr(P \llbracket (\mathbf{while})^n \rrbracket (\rho)) \\
&= tr\left(P \bigvee_{n=0}^\infty \llbracket (\mathbf{while})^n \rrbracket (\rho)\right) \\
&= tr(P \llbracket \mathbf{while } M[\bar{q}] = 1 \text{ do } S' \rrbracket (\rho)).
\end{aligned}$$

So if

$$\models_{tot} \{Q\} \mathbf{while } M[\bar{q}] = 1 \text{ do } S' \{P\},$$

then

$$tr(Q\rho) \leq tr\left(\left(\bigvee_{n=0}^\infty P_n\right)\rho\right)$$

for all ρ , and by Lemma 2.1 we obtain $Q \sqsubseteq \bigvee_{n=0}^\infty P_n$. $\square$

PROPOSITION 7.2.

- (1) (a) $wlp.\mathbf{skip}.P = P$.
- (b) If $type(q) = \mathbf{Boolean}$, then

$$wlp.(q := 0).P = |0\rangle_q \langle 0| P |0\rangle_q \langle 0| + |1\rangle_q \langle 0| P |0\rangle_q \langle 1|,$$

and if $\text{type}(q) = \mathbf{integer}$, then

$$\text{wlp}.(q := 0).P = \sum_{n=-\infty}^{\infty} |n\rangle_q \langle 0|P|0\rangle_q \langle n|.$$

- (c) $\text{wlp}.(\bar{q} := U\bar{q}).P = U^\dagger P U.$
- (d) $\text{wlp}.(S_1; S_2).P = \text{wlp}.S_1.(\text{wlp}.S_2.P).$
- (e) $\text{wlp}.(\mathbf{measure} M[\bar{q}] : \bar{S}).P = \sum_m M_m^\dagger (\text{wlp}.S_m.P) M_m.$
- (f) $\text{wlp}.(\mathbf{while} M[\bar{q}] = 1 \mathbf{do} S).P = \bigwedge_{n=0}^{\infty} P_n$, where

$$\begin{cases} P_0 = I_{\mathcal{H}_{all}}, \\ P_{n+1} = M_0^\dagger P M_0 + M_1^\dagger (\text{wlp}.S.P_n) M_1 \text{ for all } n \geq 0, \end{cases}$$

and $\bigwedge$ stands for the greatest lower bound of quantum predicates according to the Löwner partial order $\sqsubseteq$.

- (2) For any quantum program S , for any quantum predicate $P \in \mathcal{P}(\mathcal{H}_{all})$, and for any partial density operator $\rho \in \mathcal{D}^-(\mathcal{H}_{all})$, we have

$$\text{tr}((\text{wlp}.S.P)\rho) = \text{tr}(P\llbracket S \rrbracket(\rho)) + [\text{tr}(\rho) - \text{tr}(\llbracket S \rrbracket(\rho))].$$

PROOF. Similar to the case of weakest precondition, we prove (1) and (2) simultaneously by induction on the structure of quantum program S .

Case 1. $S \equiv \mathbf{skip}$, or $q := 0$, or $\bar{q} := U\bar{q}$. Similar to Cases 1, 2, and 3 in the proof of Proposition 7.1.

Case 2. $S \equiv S_1; S_2$. First, with the induction hypothesis on S_1 and S_2 , we have

$$\begin{aligned} \text{tr}(\text{wlp}.S_1.(\text{wlp}.S_2.P)\rho) &= \text{tr}(\text{wlp}.S_2.P\llbracket S_1 \rrbracket(\rho)) + [\text{tr}(\rho) - \text{tr}(\llbracket S_1 \rrbracket(\rho))] \\ &= \text{tr}(P\llbracket S_2 \rrbracket(\llbracket S_1 \rrbracket(\rho)) + [\text{tr}(\llbracket S_1 \rrbracket(\rho)) - \text{tr}(\llbracket S_2 \rrbracket(\llbracket S_1 \rrbracket(\rho)))] + [\text{tr}(\rho) - \text{tr}(\llbracket S_1 \rrbracket(\rho))]) \\ &= \text{tr}(P\llbracket S_2 \rrbracket(\llbracket S_1 \rrbracket(\rho)) + [\text{tr}(\rho) - \text{tr}(\llbracket S_2 \rrbracket(\llbracket S_1 \rrbracket(\rho)))] \\ &= \text{tr}(P\llbracket S \rrbracket(\rho)) + [\text{tr}(\rho) - \text{tr}(\llbracket S \rrbracket(\rho))]. \end{aligned}$$

If $\models_{par} \{Q\}S\{P\}$, then it holds that

$$\text{tr}(Q\rho) \leq \text{tr}(P\llbracket S \rrbracket(\rho)) + [\text{tr}(\rho) - \text{tr}(\llbracket S \rrbracket(\rho))] = \text{tr}(\text{wlp}.S_1.(\text{wlp}.S_2.P)\rho)$$

for all $\rho \in \mathcal{D}^-(\mathcal{H}_{all})$, and by Lemma 2.1 we obtain

$$Q \sqsubseteq \text{wlp}.S_1.(\text{wlp}.S_2.P).$$

Case 3. $S \equiv \mathbf{measure} M[\bar{q}] : \bar{S}$. It can be derived by induction hypothesis on all S_m that

$$\begin{aligned} \text{tr}\left(\sum_m M_m^\dagger (\text{wlp}.S_m.P) M_m \rho\right) &= \sum_m \text{tr}(M_m^\dagger (\text{wlp}.S_m.P) M_m \rho) \\ &= \sum_m \text{tr}((\text{wlp}.S_m.P) M_m \rho M_m^\dagger) \\ &= \sum_m \{\text{tr}(P\llbracket S_m \rrbracket(M_m \rho M_m^\dagger)) + [\text{tr}(M_m \rho M_m^\dagger) - \text{tr}(\llbracket S_m \rrbracket(M_m \rho M_m^\dagger))]\} \\ &= \sum_m \text{tr}(P\llbracket S_m \rrbracket(M_m \rho M_m^\dagger)) + \left[\sum_m \text{tr}(M_m \rho M_m^\dagger) - \sum_m \text{tr}(\llbracket S_m \rrbracket(M_m \rho M_m^\dagger)) \right] \\ &= \text{tr}\left(P \sum_m \llbracket S_m \rrbracket(M_m \rho M_m^\dagger)\right) + \left[\text{tr}\left(\rho \sum_m M_m^\dagger M_m\right) - \text{tr}\left(\sum_m \llbracket S_m \rrbracket(M_m \rho M_m^\dagger)\right) \right] \\ &= \text{tr}(P\llbracket S \rrbracket(\rho)) + [\text{tr}(\rho) - \text{tr}(\llbracket S \rrbracket(\rho))] \end{aligned}$$

because the measurement M satisfies the normalization condition $\sum_m M_m^\dagger M_m = I_{\mathcal{H}_Q}$. If $\models_{par} \{Q\}S\{P\}$, then for all $\rho \in \mathcal{D}^-(\mathcal{H}_{all})$, it holds that

$$tr(Q\rho) \leq tr(P\llbracket S \rrbracket(\rho)) + [tr(\rho) - tr(\llbracket S \rrbracket(\rho))] = tr\left(\sum_m M_m^\dagger(wlp.S_m.P)M_m\rho\right).$$

This, together, with Lemma 2.1, implies

$$Q \sqsubseteq \sum_m M_m^\dagger(wlp.S_m.P)M_m.$$

Case 4. $S \equiv \mathbf{while} M[\bar{q}] = 1 \mathbf{do} S'$. We first prove that

$$\begin{aligned} tr(P_n\rho) &= tr(P\llbracket (\mathbf{while})^n \rrbracket(\rho)) + [tr(\rho) - tr(\llbracket (\mathbf{while})^n \rrbracket(\rho))] \\ &= tr(\rho) - tr[(I - P)\llbracket (\mathbf{while})^n \rrbracket(\rho)] \end{aligned} \quad (8)$$

by induction on n , where I is the identity operator on $\mathcal{H}_{all}$, and $(\mathbf{while})^n$ is an abbreviation of $(\mathbf{while} M[\bar{q}] = 1 \mathbf{do} S')^n$. The case of $n = 0$ is obvious. By induction on S' and induction hypothesis on n , we observe the following:

$$\begin{aligned} tr(P_{n+1}\rho) &= tr[(M_0^\dagger P M_0)\rho + M_1^\dagger(wlp.S'.P_n)M_1\rho] \\ &= tr(M_0^\dagger P M_0\rho) + tr(M_1^\dagger(wlp.S'.P_n)M_1\rho) \\ &= tr(P M_0\rho M_0^\dagger) + tr((wlp.S'.P_n)M_1\rho M_1^\dagger) \\ &= tr(P M_0\rho M_0^\dagger) + tr(P_n\llbracket S' \rrbracket(M_1\rho M_1^\dagger)) + [tr(M_1\rho M_1^\dagger) - tr(\llbracket S' \rrbracket(M_1\rho M_1^\dagger))] \\ &= tr(P M_0\rho M_0^\dagger) + tr(P\llbracket (\mathbf{while})^n \rrbracket(\llbracket S \rrbracket(M_1\rho M_1^\dagger))) + [tr(\llbracket S \rrbracket(M_1\rho M_1^\dagger)) \\ &\quad - tr(\llbracket (\mathbf{while})^n \rrbracket(\llbracket S \rrbracket(M_1\rho M_1^\dagger)))] + [tr(M_1\rho M_1^\dagger) - tr(\llbracket S' \rrbracket(M_1\rho M_1^\dagger))] \\ &= tr(P[M_0\rho M_0^\dagger + \llbracket (\mathbf{while})^n \rrbracket(\llbracket S \rrbracket(M_1\rho M_1^\dagger))]) \\ &\quad + [tr(\rho) - tr(M_0\rho M_0^\dagger + \llbracket (\mathbf{while})^n \rrbracket(\llbracket S \rrbracket(M_1\rho M_1^\dagger)))] \\ &= tr(P\llbracket (\mathbf{while})^{n+1} \rrbracket(\rho)) + [tr(\rho) - tr(\llbracket (\mathbf{while})^{n+1} \rrbracket(\rho))]. \end{aligned}$$

This completes the proof of Eq. (8). Note that quantum predicate $P \sqsubseteq I$. Then $I - P$ is a positive operator. Since

$$\{\llbracket (\mathbf{while})^n \rrbracket(\rho)\}_{n=0}^\infty$$

is an increasing sequence (see the proof of Proposition 5.1, (6)), it can be observed from Eq. (8) that $tr(P_n\rho) \geq tr(P_{n+1}\rho)$ for any $n \geq 0$ and for any $\rho \in \mathcal{D}^-(\mathcal{H}_{all})$. This implies that $\{I - P_n\}_{n=0}^\infty$ is an increasing sequence of quantum predicates. Consequently, $\bigvee_{n=0}^\infty (I - P_n)$ exists because $(\mathcal{H}(\mathcal{P}_{all}), \sqsubseteq)$ is a complete partial order (CPO) (see D'Hondt and Panangaden [2006]; Proposition 2.3). Furthermore,

$$\bigwedge_{n=0}^\infty P_n = I - \bigvee_{n=0}^\infty (I - P_n)$$

exists too. Multiplying the two sides of the above equation by ρ yields

$$\left(\bigwedge_{n=0}^\infty P_n\right)\rho = \rho - \bigvee_{n=0}^\infty (I - P_n)\rho.$$

Then, by Proposition 5.1 (6), Eq. (8), and the continuity of the trace operator, we obtain

$$\begin{aligned}
tr \left(\left(\bigwedge_{n=0}^{\infty} P_n \right) \rho \right) &= tr[\rho - \bigvee_{n=0}^{\infty} (I - P_n) \rho] \\
&= tr(\rho) - tr[\bigvee_{n=0}^{\infty} (I - P_n) \rho] \\
&= tr(\rho) - \bigvee_{n=0}^{\infty} tr[(I - P_n) \rho] \\
&= \bigwedge_{n=0}^{\infty} \{tr(\rho) - tr[(I - P_n) \rho]\} \\
&= \bigwedge_{n=0}^{\infty} tr(P_n \rho) \\
&= \bigwedge_{n=0}^{\infty} \{tr(\rho) - tr[(I - P) \llbracket (\mathbf{while})^n \rrbracket(\rho)]\} \\
&= tr(\rho) - \bigvee_{n=0}^{\infty} tr[(I - P) \llbracket (\mathbf{while})^n \rrbracket(\rho)] \\
&= tr(\rho) - tr \left[(I - P) \bigvee_{n=0}^{\infty} \llbracket (\mathbf{while})^n \rrbracket(\rho) \right] \\
&= tr(\rho) - tr[(I - P) \llbracket S \rrbracket(\rho)] \\
&= tr(P \llbracket S \rrbracket(\rho)) + [tr(\rho) - tr(\llbracket S \rrbracket(\rho))].
\end{aligned}$$

For any $Q \in \mathcal{P}(\mathcal{H}_{all})$, $\models_{par} \{Q\}S\{P\}$ implies the following:

$$tr(Q\rho) \leq tr(P \llbracket S \rrbracket(\rho)) + [tr(\rho) - tr(\llbracket S \rrbracket(\rho))] = tr \left(\left(\bigwedge_{n=0}^{\infty} P_n \right) \rho \right)$$

for all $\rho \in \mathcal{D}^-(\mathcal{H}_{all})$. This, together with Lemma 2.1, leads to $Q \sqsubseteq \bigwedge_{n=0}^{\infty} P_n$. $\square$

We see that Propositions 7.1 and 7.2 coincide with Figures 2 and 3, respectively, in Feng et al. [2007] when only quantum variables of type **Boolean** are considered. But Figures 2 and 3 in Feng et al. [2007] are given directly as definitions, and their intuitive meanings are not clear because they do not have any connection to total and partial correctness of quantum programs. In contrast, Propositions 7.1 and 7.2 fill in such a gap, and they are derived from Definition 7.1, which is entirely given on the basis of the notions of total and partial correctness of quantum programs.

The next proposition gives a recursive characterization of the weakest precondition and weakest liberal precondition of a quantum loop, and provides a key step in the proof of completeness of quantum Floyd–Hoare logic.

PROPOSITION 7.3. *We write **while** for quantum loop “**while** $M[\bar{q}] = 1$ **do** S ”. Then, for any $P \in \mathcal{P}(\mathcal{H}_{all})$, we have*

- (1) $wp.\mathbf{while}.P = M_0^\dagger P M_0 + M_1^\dagger (wp.S.(wp.\mathbf{while}.P)) M_1$.
- (2) $wlp.\mathbf{while}.P = M_0^\dagger P M_0 + M_1^\dagger (wlp.S.(wlp.\mathbf{while}.P)) M_1$.

PROOF. We only prove (2), and the proof of (1) is similar and easier. For every $\rho \in \mathcal{D}^-(\mathcal{H}_{all})$, by Proposition 7.2(2) we observe that

$$\begin{aligned}
& tr[(M_0^\dagger P M_0 + M_1^\dagger (wlp.S.(wlp.\mathbf{while}.P)) M_1) \rho] \\
&= tr(P M_0 \rho M_0^\dagger) + tr[(wlp.S.(wlp.\mathbf{while}.P)) M_1 \rho M_1^\dagger] \\
&= tr(P M_0 \rho M_0^\dagger) + tr[(wlp.\mathbf{while}.P) \llbracket S \rrbracket (M_1 \rho M_1^\dagger)] \\
&\quad + [tr(M_1 \rho M_1^\dagger) - tr(\llbracket S \rrbracket (M_1 \rho M_1^\dagger))] \\
&= tr(P M_0 \rho M_0^\dagger) + tr[P \llbracket \mathbf{while} \rrbracket (\llbracket S \rrbracket (M_1 \rho M_1^\dagger))] + [tr(\llbracket S \rrbracket (M_1 \rho M_1^\dagger)) \\
&\quad quad - tr(\llbracket \mathbf{while} \rrbracket (\llbracket S \rrbracket (M_1 \rho M_1^\dagger))] + [tr(M_1 \rho M_1^\dagger) - tr(\llbracket S \rrbracket (M_1 \rho M_1^\dagger))] \\
&= tr[P(M_0 \rho M_0^\dagger + \llbracket \mathbf{while} \rrbracket (\llbracket S \rrbracket (M_1 \rho M_1^\dagger)))] \\
&\quad + [tr(M_1 \rho M_1^\dagger) - tr(\llbracket \mathbf{while} \rrbracket (\llbracket S \rrbracket (M_1 \rho M_1^\dagger)))] \\
&= tr(P \llbracket \mathbf{while} \rrbracket (\rho)) + [tr(\rho M_1^\dagger M_1) - tr(\llbracket \mathbf{while} \rrbracket (\llbracket S \rrbracket (M_1 \rho M_1^\dagger)))] \\
&= tr(P \llbracket \mathbf{while} \rrbracket (\rho)) + [tr(\rho(I - M_0^\dagger M_0)) - tr(\llbracket \mathbf{while} \rrbracket (\llbracket S \rrbracket (M_1 \rho M_1^\dagger)))] \\
&= tr(P \llbracket \mathbf{while} \rrbracket (\rho)) + [tr(\rho) - tr(M_0 \rho M_0^\dagger + \llbracket \mathbf{while} \rrbracket (\llbracket S \rrbracket (M_1 \rho M_1^\dagger)))] \\
&= tr(P \llbracket \mathbf{while} \rrbracket (\rho)) + [tr(\rho) - tr(\llbracket \mathbf{while} \rrbracket (\rho))].
\end{aligned}$$

This means that

$$\{M_0^\dagger P M_0 + M_1^\dagger (wlp.S.(wlp.\mathbf{while}.P)) M_1\} \mathbf{while}\{P\},$$

and

$$Q \sqsubseteq M_0^\dagger P M_0 + M_1^\dagger (wlp.S.(wlp.\mathbf{while}.P)) M_1$$

provided $\models_{par} \{Q\} \mathbf{while}\{P\}$. $\square$

8. PROOF SYSTEM FOR PARTIAL CORRECTNESS

Now we are ready to present an axiomatic system of Floyd–Hoare logic for quantum programs. The quantum Floyd–Hoare logic can be divided into two proof systems, one for partial correctness and one for total correctness. In this section, we introduce the proof system qPD for partial correctness of quantum programs. The system qPD can be seen as a quantum counterpart of the proof system PD in Apt and Olderog [1997] for partial correctness of (classical) deterministic programs. It consists of the axioms and inference rules in Figure 2.

We first prove the soundness of the proof system qPD with respect to the semantics of partial correctness: provability of a correctness formula in qPD implies its truth in the sense of partial correctness.

THEOREM 8.1 (SOUNDNESS). *The proof system qPD is sound for partial correctness of quantum programs; that is, for any quantum program S and quantum predicates $P, Q \in \mathcal{P}(\mathcal{H}_{all})$, we have*

$$\vdash_{qPD} \{P\} S \{Q\} \text{ implies } \models_{par} \{P\} S \{Q\}.$$

PROOF. We only need to show that the axioms and inference rules of qPD are valid in the sense of partial correctness.

(*Axiom Skip*). It is obvious that $\models_{par} \{P\} \mathbf{skip} \{P\}$.

(*Axiom Initialization*). We only prove the case of $type(q) = \mathbf{integer}$; the case of $type(q) = \mathbf{Boolean}$ is similar. For any $\rho \in \mathcal{D}^-(\mathcal{H}_{all})$, it follows from Proposition 5.1.2

$$(Axiom\ Skip) \quad \{P\} \mathbf{Skip} \{P\}$$

(Axiom Initialization) (1) If $type(q) = \mathbf{Boolean}$, then

$$\{|0\rangle_q \langle 0| P |0\rangle_q \langle 0| + |1\rangle_q \langle 0| P |0\rangle_q \langle 1|\} q := 0 \{P\}$$

(2) If $type(q) = \mathbf{integer}$, then

$$\left\{ \sum_{n=-\infty}^{\infty} |n\rangle_q \langle 0| P |0\rangle_q \langle n| \right\} q := 0 \{P\}$$

$$(Axiom\ Unitary\ Transformation) \quad \{U^\dagger P U\} \bar{q} := U \bar{q} \{P\}$$

$$(Rule\ Sequential\ Composition) \quad \frac{\{P\} S_1 \{Q\} \quad \{Q\} S_2 \{R\}}{\{P\} S_1; S_2 \{R\}}$$

$$(Rule\ Measurement) \quad \frac{\{P_m\} S_m \{Q\} \text{ for all } m}{\{\sum_m M_m^\dagger P_m M_m\} \mathbf{measure} M[\bar{q}] : \bar{S}\{Q\}}$$

$$(Rule\ Loop\ Partial) \quad \frac{\{Q\} S \{M_0^\dagger P M_0 + M_1^\dagger Q M_1\}}{\{M_0^\dagger P M_0 + M_1^\dagger Q M_1\} \mathbf{while} M[\bar{q}] = 1 \mathbf{do} S \{P\}}$$

$$(Rule\ Order) \quad \frac{P \sqsubseteq P' \quad \{P'\} S \{Q'\} \quad Q' \sqsubseteq Q}{\{P\} S \{Q\}}$$

Fig. 2. Proof system qPD of partial correctness.

that

$$\begin{aligned} tr \left[\left(\sum_{n=-\infty}^{\infty} |n\rangle_q \langle 0| P |0\rangle_q \langle n| \right) \rho \right] &= \sum_{n=-\infty}^{\infty} tr(|n\rangle_q \langle 0| P |0\rangle_q \langle n| \rho) \\ &= \sum_{n=-\infty}^{\infty} tr(P |0\rangle_q \langle n| \rho |n\rangle_q \langle 0|) \\ &= tr \left(P \sum_{n=-\infty}^{\infty} |0\rangle_q \langle n| \rho |n\rangle_q \langle 0| \right) \\ &= tr(P \llbracket q := 0 \rrbracket (\rho)). \end{aligned}$$

Therefore, we have

$$\models_{par} \left\{ \sum_{n=-\infty}^{\infty} |n\rangle_q \langle 0| P |0\rangle_q \langle n| \right\} q := 0 \{P\}.$$

(*Axiom Unitary Transformation*). It is easy to see that

$$\models_{par} \{U^\dagger P U\} \bar{q} := U \bar{q} \{P\}.$$

(*Rule Sequential Composition*). If $\models_{par} \{P\} S_1 \{Q\}$ and $\models_{par} \{Q\} S_2 \{R\}$, then for any $\rho \in \mathcal{D}^-(\mathcal{H}_{all})$ we have

$$\begin{aligned} tr(P\rho) &\leq tr(Q\llbracket S_1 \rrbracket(\rho)) + [tr(\rho) - tr(\llbracket S_1 \rrbracket(\rho))] \\ &\leq tr(R\llbracket S_2 \rrbracket(\llbracket S_1 \rrbracket(\rho))) + [tr(\llbracket S_1 \rrbracket(\rho)) - tr(\llbracket S_2 \rrbracket(\llbracket S_1 \rrbracket(\rho)))] \\ &\quad + [tr(\rho) - tr(\llbracket S_1 \rrbracket(\rho))] \\ &= tr(R\llbracket S_1; S_2 \rrbracket(\rho)) + [tr(\rho) - tr(\llbracket S_1; S_2 \rrbracket(\rho))]. \end{aligned}$$

Therefore, $\models_{par} \{P\} S_1; S_2 \{R\}$, as desired.

(*Rule Measurement*). Assume that $\models_{par} \{P_m\} S_m \{Q\}$ for all m . Then, for all $\rho \in \mathcal{D}^-(\mathcal{H}_{all})$, since $\sum_m M_m^\dagger M_m = I_{\mathcal{H}_{\bar{q}}}$, it holds that

$$\begin{aligned} tr\left(\sum_m M_m^\dagger P_m M_m \rho\right) &= \sum_m tr(M_m^\dagger P_m M_m \rho) \\ &= \sum_m tr(P_m M_m \rho M_m^\dagger) \\ &\leq \sum_m \{tr(Q\llbracket S_m \rrbracket(M_m \rho M_m^\dagger)) + [tr(M_m \rho M_m^\dagger) - tr(\llbracket S_m \rrbracket(M_m \rho M_m^\dagger))]\} \\ &= \sum_m tr(Q\llbracket S_m \rrbracket(M_m \rho M_m^\dagger)) + \left[\sum_m tr(M_m \rho M_m^\dagger) - \sum_m tr(\llbracket S_m \rrbracket(M_m \rho M_m^\dagger))\right] \\ &= tr(Q \sum_m \llbracket S_m \rrbracket(M_m \rho M_m^\dagger)) + [tr(\sum_m \rho M_m^\dagger M_m) - tr(\sum_m \llbracket S_m \rrbracket(M_m \rho M_m^\dagger))] \\ &= tr(Q\llbracket \mathbf{measure} \rrbracket(\rho)) + [tr(\rho) - tr(\llbracket \mathbf{measure} \rrbracket(\rho))], \end{aligned}$$

and

$$\models_{par} \left\{ \sum_m M_m^\dagger P_m M_m \right\} \mathbf{measure}\{Q\},$$

where **measure** is an abbreviation of statement “**measure** $M[\bar{q}] : \bar{S}$ ”.

(*Rule Loop Partial*). Suppose that

$$\models_{par} \{Q\} S \{M_0^\dagger P M_0 + M_1^\dagger Q M_1\}.$$

Then, for all $\rho \in \mathcal{D}^-(\mathcal{H}_{all})$, it holds that

$$tr(Q\rho) \leq tr((M_0^\dagger P M_0 + M_1^\dagger Q M_1)\llbracket S \rrbracket(\rho)) + [tr(\rho) - tr(\llbracket S \rrbracket(\rho))]. \quad (9)$$

Furthermore, we have

$$\begin{aligned} tr[(M_0^\dagger P M_0 + M_1^\dagger Q M_1)\rho] &\leq \sum_{k=0}^n tr(P(\mathcal{E}_0 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^k)(\rho)) \\ &\quad + tr(Q(\mathcal{E}_1 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^n)(\rho)) \\ &\quad + \sum_{k=0}^{n-1} [tr(\mathcal{E}_1 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^k(\rho)) - tr((\llbracket S \rrbracket \circ \mathcal{E}_1)^{k+1}(\rho))] \end{aligned} \quad (10)$$

for all $n \geq 1$. In fact, Eq. (10) may be proved by induction on n . The case of $n = 1$ is obvious. Using Eq. (9), we obtain the following:

$$\begin{aligned}
 tr(Q(\mathcal{E}_1 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^n)(\rho)) &\leq tr((M_0^\dagger P M_0 + M_1^\dagger Q M_1)(\llbracket S \rrbracket \circ \mathcal{E}_1)^{n+1}(\rho)) \\
 &\quad + [tr((\mathcal{E}_1 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^n)(\rho)) - tr((\llbracket S \rrbracket \circ \mathcal{E}_1)^{n+1}(\rho))] \\
 &= tr(P(\mathcal{E}_0 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^{n+1})(\rho)) + tr(Q(\mathcal{E}_1 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^{n+1})(\rho)) \\
 &\quad + [tr((\mathcal{E}_1 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^n)(\rho)) - tr((\llbracket S \rrbracket \circ \mathcal{E}_1)^{n+1}(\rho))].
 \end{aligned} \tag{11}$$

Combining Eqs. (10) and (11), we assert that

$$\begin{aligned}
 tr[(M_0^\dagger P M_0 + M_1^\dagger Q M_1)\rho] &\leq \sum_{k=0}^{n+1} tr(P(\mathcal{E}_0 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^k)(\rho)) \\
 &\quad + tr(Q(\mathcal{E}_1 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^{n+1})(\rho)) \\
 &\quad + \sum_{k=0}^n [tr(\mathcal{E}_1 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^k(\rho)) - tr((\llbracket S \rrbracket \circ \mathcal{E}_1)^{k+1}(\rho))].
 \end{aligned}$$

Therefore, Eq. (10) holds in the case of $n + 1$, provided it is true in the case of n , and we complete the proof of Eq. (10).

Now we note that

$$\begin{aligned}
 tr((\mathcal{E}_1 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^k)(\rho)) &= tr(M_1(\llbracket S \rrbracket \circ \mathcal{E}_1)^k(\rho)M_1^\dagger) \\
 &= tr((\llbracket S \rrbracket \circ \mathcal{E}_1)^k(\rho)M_1^\dagger M_1) \\
 &= tr((\llbracket S \rrbracket \circ \mathcal{E}_1)^k(\rho)(I - M_0^\dagger M_0)) \\
 &= tr((\llbracket S \rrbracket \circ \mathcal{E}_1)^k(\rho)) - tr((\mathcal{E}_0 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^k)(\rho))
 \end{aligned}$$

because the yes-no measurement $M = \{M_0, M_1\}$ satisfies the normalization condition $M_0^\dagger M_0 + M_1^\dagger M_1 = I$, where I is the identity operator on $\mathcal{H}_{all}$. Then, it follows that

$$\begin{aligned}
 \sum_{k=0}^{n-1} [tr(\mathcal{E}_1 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^k(\rho)) - tr((\llbracket S \rrbracket \circ \mathcal{E}_1)^{k+1}(\rho))] &= \sum_{k=0}^{n-1} tr((\llbracket S \rrbracket \circ \mathcal{E}_1)^k(\rho)) \\
 - \sum_{k=0}^{n-1} [tr(\mathcal{E}_0 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^k(\rho)) - tr((\llbracket S \rrbracket \circ \mathcal{E}_1)^{k+1}(\rho))] &\tag{12} \\
 = tr(\rho) - tr((\llbracket S \rrbracket \circ \mathcal{E}_1)^n(\rho)) - \sum_{k=0}^{n-1} tr(\mathcal{E}_0 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^k(\rho)).
 \end{aligned}$$

On the other hand, we have

$$\begin{aligned}
 tr(Q(\mathcal{E}_1 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^n)(\rho)) &= tr(Q M_1(\llbracket S \rrbracket \circ \mathcal{E}_1)^n(\rho)M_1^\dagger) \\
 &\leq tr(M_1(\llbracket S \rrbracket \circ \mathcal{E}_1)^n(\rho)M_1^\dagger) \\
 &= tr((\llbracket S \rrbracket \circ \mathcal{E}_1)^n(\rho)M_1^\dagger M_1) \\
 &= tr((\llbracket S \rrbracket \circ \mathcal{E}_1)^n(\rho)(I - M_0^\dagger M_0)) \\
 &= tr((\llbracket S \rrbracket \circ \mathcal{E}_1)^n(\rho)) - tr((\mathcal{E}_0 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^n)(\rho)).
 \end{aligned} \tag{13}$$

Putting Eqs. (12) and (13) into Eq. (10), we obtain

$$\begin{aligned}
tr[(M_0^\dagger P M_0 + M_1^\dagger Q M_1)\rho] &\leq \sum_{k=0}^n tr(P(\mathcal{E}_0 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^k)(\rho)) \\
&\quad + \left[tr(\rho) - \sum_{k=0}^n tr((\mathcal{E}_0 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^k)(\rho)) \right] \\
&= tr \left(P \sum_{k=0}^n (\mathcal{E}_0 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^k)(\rho) \right) \\
&\quad + \left[tr(\rho) - tr \left(\sum_{k=0}^n (\mathcal{E}_0 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^k)(\rho) \right) \right].
\end{aligned}$$

Let $n \rightarrow \infty$. Then it follows that

$$tr[(M_0^\dagger P M_0 + M_1^\dagger Q M_1)\rho] \leq tr(P \llbracket \mathbf{while} \rrbracket(\rho)) + [tr(\rho) - tr(\llbracket \mathbf{while} \rrbracket(\rho))]$$

and

$$\models_{par} \{M_0^\dagger P M_0 + M_1^\dagger Q M_1\} \mathbf{while}\{P\},$$

where **while** is an abbreviation of quantum loop “**while** $M[\bar{q}] = 1$ **do** S ”. $\square$

Now we will establish completeness for the proof system qPD with respect to the semantics of partial correctness: truth of a quantum program in the sense of partial correctness implies its provability in qPD . Note that the Löwner ordering assertions between quantum predicates in (Rule order) are statements about complex numbers. So only a completeness of qPD relative to the theory of the field of complex numbers may be anticipated; more precisely, we can add all statements that are true in the field of complex numbers into qPD in order to make it complete. The following theorem should be understood exactly in the sense of such a relative completeness.

THEOREM 8.2 (COMPLETENESS). *The proof system qPD is complete for partial correctness of quantum programs; that is, for any quantum program S and quantum predicates $P, Q \in \mathcal{P}(\mathcal{H}_{all})$, we have*

$$\models_{par} \{P\}S\{Q\} \text{ implies } \vdash_{qPD} \{P\}S\{Q\}.$$

PROOF. If $\models_{par} \{P\}S\{Q\}$, then by Definition 7.1, (2) we have $P \sqsubseteq wlp.S.Q$. Therefore, by (Rule order) it suffices to prove the following:

— *Claim:* $\vdash_{qPD} \{wlp.S.Q\}S\{Q\}$.

We proceed by induction on the structure of S .

Case 1. $S \equiv \mathbf{skip}$. Immediate from (Axiom Skip).

Case 2. $S \equiv q := 0$. Immediate from (Axiom Initialization).

Case 3. $S \equiv \bar{q} := U\bar{q}$. Immediate from (Axiom Unitary Transformation).

Case 4. $S \equiv S_1; S_2$. It follows from the induction hypothesis on S_1 and S_2 that

$$\vdash_{qPD} \{wlp.S_1.(wlp.S_2.Q)\}S_1\{wlp.S_2.Q\}$$

and

$$\vdash_{qPD} \{wlp.S_2.Q\}S_2\{Q\}.$$

We obtain the following:

$$\vdash_{qPD} \{wlp.S_1.(wlp.S_2.Q)\}S_1; S_2\{Q\}$$

by (*Rule Sequential Composition*). Then with Proposition 7.2, (1.d) we see that

$$\vdash_{qPD} \{wlp.S_1; S_2.Q\}S_1; S_2\{Q\}.$$

Case 5. $S \equiv \mathbf{measure} M[\bar{q}] : \bar{S}$. For all m , by the induction hypothesis on S_m we obtain

$$\vdash_{qPD} \{wlp.S_m.Q\}S_m\{Q\}.$$

Then applying (*Rule Measurement*) yields

$$\vdash_{qPD} \left\{ \sum_m M_m^\dagger (wlp.S_m.Q) M_m \right\} \mathbf{measure} M[\bar{q}] : \bar{S}\{Q\},$$

and using Proposition 7.2, (1.e) we have

$$\vdash_{qPD} \{wlp.\mathbf{measure} M[\bar{q}] : \bar{S}.P\} \mathbf{measure} M[\bar{q}] : \bar{S}\{Q\}.$$

Case 6. $S \equiv \mathbf{while} M[\bar{q}] = 1 \mathbf{do} S'$. For simplicity, we write **while** for quantum loop “**while** $M[\bar{q}] = 1 \mathbf{do} S'$ ”. The induction hypothesis on S asserts that

$$\vdash_{qPD} \{wlp.S.(wlp.\mathbf{while}.P)\}S\{wlp.\mathbf{while}.P\}.$$

By Proposition 7.3,(2) we have

$$wlp.\mathbf{while}.P = M_0^\dagger P M_0 + M_1^\dagger (wlp.S.(wlp.\mathbf{while}.P)) M_1.$$

Then, by (*Rule Loop Par*), we obtain

$$\vdash_{qPD} \{wlp.\mathbf{while}.P\} \mathbf{while}\{P\},$$

as desired. $\square$

9. PROOF SYSTEM FOR TOTAL CORRECTNESS

The aim of this section is to present a proof system qTD for total correctness of quantum program. The system qTD can be seen as a quantum generalization of the proof system TD given in Apt and Olderog [1997] for total correctness of (classical) deterministic programs.

The only difference between qTD and qPD is the inference rule for quantum loops. To give the rule for total correctness of quantum loops, we need a notion of bound function, which expresses the number of iterations of a quantum loop in its computation.

Definition 9.1. Let $P \in \mathcal{P}(\mathcal{H}_{all})$ and $\epsilon > 0$. A function $t : \mathcal{D}^-(\mathcal{H}_{all}) \rightarrow \mathbb{N} \cup \{\infty\}$ is called a (P, ϵ) –bound function of quantum loop “**while** $M[\bar{q}] = 1 \mathbf{do} S$ ” if it satisfies the following conditions:

- (1) $t(\llbracket S \rrbracket(M_1 \rho M_1^\dagger)) \leq t(\rho)$; and
- (2) $tr(P\rho) \geq \epsilon$ implies $t(\llbracket S \rrbracket(M_1 \rho M_1^\dagger)) < t(\rho)$

for all $\rho \in \mathcal{D}^-(\mathcal{H}_{all})$, where $\mathbb{N}$ is the set of nonnegative integers.

Recall that a bound function t of a classical loop “**while** $B \mathbf{do} S$ ” satisfies the inequality $t(\llbracket S \rrbracket(s)) < t(s)$ for any input state s . It is interesting to compare this inequality with conditions (1) and (2) of the above definition, and we see that the latter are two inequalities between $t(\llbracket S \rrbracket(M_1 \rho M_1^\dagger))$ and $t(\rho)$, but not between $t(\llbracket S \rrbracket(\rho))$ and $t(\rho)$. This is because in the implementation of the quantum loop “**while** $M[\bar{q}] = 1 \mathbf{do} S$ ”, we need to perform the yes-no measurement M on ρ when checking the loop guard “ $M[\bar{q}] = 1$ ”, and the state of quantum variables will become $M_1 \rho M_1^\dagger$ from ρ whence the measurement outcome “yes” is observed.

The following lemma gives a characterization of the existence of bound function of a quantum loop in terms of the limit of the state of quantum variables when the number of iterations of the loop goes to infinity. It provides a key step for the proof of soundness and completeness of the proof system qTD .

LEMMA 9.1. *Let $P \in \mathcal{P}(\mathcal{H}_{all})$. Then the following two statements are equivalent:*

- (1) *for any $\epsilon > 0$, there exists a (P, ϵ) -bound function t_ϵ of quantum loop “**while** $M[\bar{q}] = 1$ **do** S ”;*
- (2) *$\lim_{n \rightarrow \infty} tr(P(\llbracket S \rrbracket \circ \mathcal{E}_1)^n(\rho)) = 0$ for all $\rho \in \mathcal{D}^-(\mathcal{H}_{all})$.*

PROOF. (1 $\Rightarrow$ 2) We prove this by contradiction. If

$$\lim_{n \rightarrow \infty} tr(P(\llbracket S \rrbracket \circ \mathcal{E}_1)^n(\rho)) \neq 0,$$

then there exist $\epsilon_0 > 0$ and strictly increasing sequence $\{n_k\}$ of nonnegative integers such that

$$tr(P(\llbracket S \rrbracket \circ \mathcal{E}_1)^{n_k}(\rho)) \geq \epsilon_0$$

for all $k \geq 0$. Thus, we have a (P, ϵ_0) -bound function of loop “**while** $M[\bar{q}] = 1$ **do** S ”. For each $k \geq 0$, we set

$$\rho_k = (\llbracket S \rrbracket \circ \mathcal{E}_1)^{n_k}(\rho).$$

Then it holds that $tr(P\rho_k) \geq \epsilon_0$, and by conditions (1) and (2) in Definition 9.1, we obtain

$$\begin{aligned} t_{\epsilon_0}(\rho_k) &> t_{\epsilon_0}(\llbracket S \rrbracket(M_1\rho_k M_1^\dagger)) \\ &= t_{\epsilon_0}((\llbracket S \rrbracket \circ \mathcal{E}_1)(\rho_k)) \\ &\geq t_{\epsilon_0}((\llbracket S \rrbracket \circ \mathcal{E}_1)^{n_{k+1}-n_k}(\rho_k)) \\ &= t_{\epsilon_0}(\rho_{k+1}). \end{aligned}$$

Consequently, we have an infinitely descending chain $\{t_{\epsilon_0}(\rho_k)\}$ in the set $\mathbb{N}$ of nonnegative integers. This is a contradiction because $\mathbb{N}$ is a well-founded set.

(2 $\Rightarrow$ 1) For each $\rho \in \mathcal{D}^-(\mathcal{H}_{all})$, if

$$\lim_{n \rightarrow \infty} tr(P(\llbracket S \rrbracket \circ \mathcal{E}_1)^n(\rho)) = 0,$$

then for any $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$tr(P(\llbracket S \rrbracket \circ \mathcal{E}_1)^n(\rho)) < \epsilon$$

for all $n \geq N$. We define the following:

$$t_\epsilon(\rho) = \min\{N \in \mathbb{N} : tr(P(\llbracket S \rrbracket \circ \mathcal{E}_1)^n(\rho)) < \epsilon \text{ for all } n \geq N\}.$$

Now it suffices to show that t_ϵ is a (P, ϵ) -bound function of loop “**while** $M[\bar{q}] = 1$ **do** S ”. To this end, we consider the following two cases:

Case 1. $tr(P\rho) \geq \epsilon$. Suppose that $t_\epsilon(\rho) = N$. Then $tr(P\rho) \geq \epsilon$ implies $N \geq 1$. By the definition of t_ϵ , we assert that

$$tr(P(\llbracket S \rrbracket \circ \mathcal{E}_1)^n(\rho)) < \epsilon$$

for all $n \geq N$. Thus, for all $n \geq N - 1 \geq 0$,

$$tr(P(\llbracket S \rrbracket \circ \mathcal{E}_1)^n(\llbracket S \rrbracket(M_1^\dagger\rho M_1))) = tr(P(\llbracket S \rrbracket \circ \mathcal{E}_1)^{n+1}(\rho)) < \epsilon.$$

Therefore,

$$t_\epsilon(\llbracket S \rrbracket(M_1^\dagger\rho M_1)) \leq N - 1 < N = t_\epsilon(\rho).$$

$$\begin{array}{c}
\{Q\}S\{M_0^\dagger PM_0 + M_1^\dagger QM_1\} \\
\text{for any } \epsilon > 0, t_\epsilon \text{ is a } (M_1^\dagger QM_1, \epsilon) - \text{bound function} \\
\text{of loop } \mathbf{while } M[\bar{q}] = 1 \mathbf{do } S \\
\hline
\{M_0^\dagger PM_0 + M_1^\dagger QM_1\} \mathbf{while } M[\bar{q}] = 1 \mathbf{do } S\{P\}
\end{array}$$

(Rule Loop Total)

Fig. 3. Proof system qTD of total correctness.

Case 2. $tr(P\rho) < \epsilon$. Again, suppose that $t_\epsilon(\rho) = N$. Now we have the following two subcases:

Subcase 2.1. $N = 0$. Then, for all $n \geq 0$,

$$tr(P(\llbracket S \rrbracket \circ \mathcal{E}_1)^n(\rho)) < \epsilon.$$

It is easy to see that $t_\epsilon(\llbracket S \rrbracket(M_1\rho M_1^\dagger)) = 0 = t_\epsilon(\rho)$.

Subcase 2.2. $N \geq 1$. We can derive that

$$t_\epsilon(\rho) > t_\epsilon(\llbracket S \rrbracket(M_1\rho M_1^\dagger))$$

in the manner of Case 1. $\square$

Now we are ready to present the proof system qTD for total correctness of quantum programs. It consists of the axioms (Axiom Skip), (Axiom Initialization), and (Axiom Unitary Transformation) and inference rules (Rule Sequential Composition), (Rule Measurement), and (Rule Order) in Figure 2 as well as inference rule (Rule Loop Total) in Figure 3.

The remainder of this section is devoted to establishing the soundness and completeness of qTD : provability of a correctness formula in qTD is equivalent to its truth in the sense of total correctness.

THEOREM 9.1 (SOUNDNESS). *The proof system qTD is sound for total correctness of quantum programs; that is, for any quantum program S and quantum predicates $P, Q \in \mathcal{P}(\mathcal{H}_{all})$, we have*

$$\vdash_{qTD} \{P\}S\{Q\} \text{ implies } \models_{tot} \{P\}S\{Q\}.$$

PROOF. It suffices to show that the axioms and inference rules of qTD are valid in the sense of total correctness.

The proof for soundness of (Axiom Skip), (Axiom Initialization), and (Axiom Unitary Transformation) is similar to the case of partial correctness.

(Rule Sequential Composition). Suppose that $\models_{tot} \{P\}S_1\{Q\}$ and $\models_{tot} \{Q\}S_2\{R\}$. Then, for any $\rho \in \mathcal{D}^-(\mathcal{H}_{all})$, with Proposition 5.1.4 we obtain

$$\begin{aligned}
tr(P\rho) &\leq tr(Q\llbracket S_1 \rrbracket(\rho)) \\
&\leq tr(R\llbracket S_2 \rrbracket(\llbracket S_1 \rrbracket(\rho))) \\
&= tr(P\llbracket S_1; S_2 \rrbracket(\rho)).
\end{aligned}$$

Therefore, $\models_{tot} \{P\}S_1; S_2\{R\}$.

(Rule Measurement). Suppose that $\models_{tot} \{P_m\}S_m\{Q\}$ for all m . Then, for any $\rho \in \mathcal{D}^-(\mathcal{H}_{all})$, it holds that

$$tr(P_m M_m \rho M_m^\dagger) \leq tr(Q\llbracket S_m \rrbracket(M_m \rho M_m^\dagger))$$

because $\{P_m\}S_m\{Q\}$ for all m . Therefore, we have

$$\begin{aligned} \text{tr} \left(\sum_m M_m^\dagger P_m M_m \rho \right) &= \sum_m \text{tr}(P_m M_m \rho M_m^\dagger) \\ &\leq \sum_m \text{tr}(Q \llbracket S_m \rrbracket (M_m \rho M_m^\dagger)) \\ &= \text{tr}(Q \sum_m \llbracket S_m \rrbracket (M_m \rho M_m^\dagger)) \\ &= \text{tr}(Q \llbracket \mathbf{measure} M[\bar{q}] : \bar{S} \rrbracket (\rho)), \end{aligned}$$

and

$$\models_{\text{tot}} \left\{ \sum_m M_m^\dagger P_m M_m \right\} \mathbf{measure} M[\bar{q}] : \bar{S} \{Q\}.$$

(Rule Loop Total). If

$$\models_{\text{tot}} \{Q\} S \{M_0^\dagger P M_0 + M_1^\dagger Q M_1\},$$

then for any $\rho \in \mathcal{D}^-(\mathcal{H}_{\text{all}})$, we have

$$\text{tr}(Q\rho) \leq \text{tr}((M_0^\dagger P M_0 + M_1^\dagger Q M_1) \llbracket S \rrbracket (\rho)). \quad (14)$$

We first prove the following inequality:

$$\begin{aligned} &\text{tr}[(M_0^\dagger P M_0 + M_1^\dagger Q M_1)\rho] \\ &\leq \sum_{k=0}^n \text{tr}(P[\mathcal{E}_0 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)]^k(\rho)) + \text{tr}(Q[\mathcal{E}_1 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^n](\rho)) \end{aligned} \quad (15)$$

by induction on n . It holds that

$$\begin{aligned} \text{tr}[(M_0^\dagger P M_0 + M_1^\dagger Q M_1)\rho] &= \text{tr}(P M_0 \rho M_0^\dagger) + \text{tr}(Q M_1 \rho M_1^\dagger) \\ &= \text{tr}(P \mathcal{E}_0(\rho)) + \text{tr}(Q \mathcal{E}_1(\rho)). \end{aligned}$$

So Eq. (15) is correct for the base case of $n = 0$. Assume Eq. (15) is correct for the case of $n = m$. Then, applying Eq. (14), we obtain the following:

$$\begin{aligned} \text{tr}[(M_0^\dagger P M_0 + M_1^\dagger Q M_1)\rho] &= \text{tr}(P \mathcal{E}_0(\rho)) + \text{tr}(Q M_1 \rho M_1^\dagger) \\ &\leq \sum_{k=0}^m \text{tr}(P[\mathcal{E}_0 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)]^k(\rho)) + \text{tr}(Q[\mathcal{E}_1 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^m](\rho)) \\ &\leq \sum_{k=0}^m \text{tr}(P[\mathcal{E}_0 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)]^k(\rho)) + \text{tr}((M_0^\dagger P M_0 + M_1^\dagger Q M_1) \llbracket S \rrbracket ([\mathcal{E}_1 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^m](\rho))) \\ &= \sum_{k=0}^m \text{tr}(P[\mathcal{E}_0 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)]^k(\rho)) + \text{tr}(P M_0 \llbracket S \rrbracket ([\mathcal{E}_1 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^m](\rho)) M_0^\dagger) \\ &\quad + \text{tr}(Q M_1 \llbracket S \rrbracket ([\mathcal{E}_1 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^m](\rho)) M_1^\dagger) \\ &= \sum_{k=0}^{m+1} \text{tr}(P[\mathcal{E}_0 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)]^k(\rho)) + \text{tr}(Q[\mathcal{E}_1 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^{m+1}](\rho)). \end{aligned}$$

Therefore, Eq. (15) also holds for the case of $n = M + 1$. Now, since for any $\epsilon > 0$ there exists $M_1^\dagger Q M_1, \epsilon$ -bound function t_ϵ of quantum loop “**while** $M[\bar{q}] = 1$ **do** S ”, by Lemma 9.1, we have

$$\begin{aligned} \lim_{n \rightarrow \infty} \text{tr}(Q[\mathcal{E}_1 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^n](\rho)) &= \lim_{n \rightarrow \infty} \text{tr}(Q M_1^\dagger (\llbracket S \rrbracket \circ \mathcal{E}_1)^n(\rho) M_1) \\ &= \lim_{n \rightarrow \infty} \text{tr}(M_1^\dagger Q M_1 (\llbracket S \rrbracket \circ \mathcal{E}_1)^n(\rho)) = 0. \end{aligned}$$

Consequently, it holds that

$$\begin{aligned} \text{tr}[(M_0^\dagger P M_0 + M_1^\dagger Q M_1)\rho] &\leq \lim_{n \rightarrow \infty} \sum_{k=0}^n \text{tr}(P[\mathcal{E}_0 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^k](\rho)) \\ &\quad + \lim_{n \rightarrow \infty} \text{tr}(Q[\mathcal{E}_1 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^n](\rho)) \\ &= \sum_{n=0}^{\infty} \text{tr}(P[\mathcal{E}_0 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^n](\rho)) \\ &= \text{tr}(P \sum_{n=0}^{\infty} [\mathcal{E}_0 \circ (\llbracket S \rrbracket \circ \mathcal{E}_1)^n](\rho)) \\ &= \text{tr}(P \llbracket \text{while } M[\bar{q}] = 1 \text{ do } S \rrbracket(\rho)). \end{aligned}$$

□

THEOREM 9.2 (COMPLETENESS). *The proof system qTD is complete for total correctness of quantum programs; that is, for any quantum program S and quantum predicates $P, Q \in \mathcal{P}(\mathcal{H}_{all})$, we have*

$$\models_{tot} \{P\}S\{Q\} \text{ implies } \vdash_{qTD} \{P\}S\{Q\}.$$

PROOF. Similar to the case of partial correctness, it suffices to prove the following:

— *Claim:* $\vdash_{qTD} \{wp.S.Q\}S\{Q\}$ for any quantum program S and quantum predicate $P \in \mathcal{P}(\mathcal{H}_{all})$

because by Definition 7.1, (1) we have $P \sqsubseteq wp.S.Q$ when $\models_{tot} \{P\}S\{Q\}$. The above claim can be done by induction on the structure of S . We only consider the case of $S \equiv \text{while } M[\bar{q}] = 1 \text{ do } S'$; the other cases are similar to the proof of Theorem 8.2. We write **while** for quantum loop “**while** $M[\bar{q}] = 1$ **do** S' ”. It follows from Proposition 7.3(1) that

$$wp.\text{while}.Q = M_0^\dagger Q M_0 + M_1^\dagger (wp.S.(wp.\text{while}.Q)) M_1.$$

So our aim is to derive that

$$\vdash_{qTD} \{M_0^\dagger Q M_0 + M_1^\dagger (wp.S.(wp.\text{while}.Q)) M_1\} \text{while}\{Q\}.$$

By the induction hypothesis on S' , we get

$$\vdash_{qTD} \{wp.S'.(wp.\text{while}.Q)\}S\{wp.\text{while}.Q\}.$$

Then by (Rule Loop Total) it suffices to show that, for any $\epsilon > 0$, there exists a $(M_1^\dagger (wp.S'.(wp.S.Q)) M_1, \epsilon)$ -bound function of quantum loop **while**. Applying Lemma 9.1, we only need to prove that

$$\lim_{n \rightarrow \infty} \text{tr}(M_1^\dagger (wp.S'.(wp.\text{while}.Q)) M_1 (\llbracket S' \rrbracket \circ \mathcal{E}_1)^n(\rho)) = 0. \quad (16)$$

First, by Propositions 7.1, (2) and 5.1, (6) we observe that

$$\begin{aligned}
 & tr(M_1^\dagger(wp.S'.(wp.\mathbf{while}.Q))M_1(\llbracket S' \rrbracket \circ \mathcal{E}_1)^n(\rho)) \\
 &= tr(wp.S'.(wp.\mathbf{while}.Q)M_1(\llbracket S' \rrbracket \circ \mathcal{E}_1)^n(\rho)M_1^\dagger) \\
 &= tr(wp.\mathbf{while}.Q\llbracket S' \rrbracket(M_1(\llbracket S' \rrbracket \circ \mathcal{E}_1)^n(\rho)M_1^\dagger)) \\
 &= tr(wp.\mathbf{while}.Q(\llbracket S' \rrbracket \circ \mathcal{E}_1)^{n+1}(\rho)) \\
 &= tr(Q\llbracket \mathbf{while} \rrbracket(\llbracket S' \rrbracket \circ \mathcal{E}_1)^{n+1}(\rho)) \\
 &= \sum_{k=n+1}^{\infty} tr(Q\mathcal{E}_0 \circ (\llbracket S' \rrbracket \circ \mathcal{E}_1)^k(\rho)).
 \end{aligned} \tag{17}$$

Second, we consider the following infinite series of nonnegative real numbers:

$$\sum_{n=0}^{\infty} tr(Q\mathcal{E}_0 \circ (\llbracket S' \rrbracket \circ \mathcal{E}_1)^k(\rho)) = tr\left(Q \sum_{n=0}^{\infty} [\mathcal{E}_0 \circ (\llbracket S' \rrbracket \circ \mathcal{E}_1)^k](\rho)\right). \tag{18}$$

Since $Q \subseteq I_{\mathcal{H}_{all}}$, it follows from Propositions 5.1, (6) and 5.2 that

$$\begin{aligned}
 tr\left(Q \sum_{n=0}^{\infty} [\mathcal{E}_0 \circ (\llbracket S' \rrbracket \circ \mathcal{E}_1)^k](\rho)\right) &= tr(Q\llbracket \mathbf{while} \rrbracket(\rho)) \\
 &\leq tr(\llbracket \mathbf{while} \rrbracket(\rho)) \leq tr(\rho) \leq 1.
 \end{aligned}$$

Therefore, the infinite series Eq. (18) converges. Note that Eq. (17) is the sum of the remaining terms of the infinite series Eq. (18) after the n th term. Then convergence of the infinite series Eq. (18) implies Eq. (16), and we complete the proof. $\square$

As in the case of qPD for partial correctness, it should be pointed out that the above theorem is merely a relative completeness of qTD with respect to the theory of the fields of complex numbers because (Rule Order) is also employed in qTD . Moreover, the condition about the existence of bound functions in (Rule Loop Total) even contains an asymptotic statement in which ϵ is required to range over all positive real numbers. In fact, it can be seen from the proof of Lemma 9.1 that ϵ only needs to range over a set of real numbers with 0 as its accumulation point, for example, $\{\frac{1}{n} : n = 1, 2, \dots\}$.

10. AN ILLUSTRATIVE EXAMPLE: REASONING ABOUT THE GROVER ALGORITHM

The purpose of this section is to show the effectiveness of the proof systems presented in Sections 8 and 9 by employing them to verify the correctness of the Grover quantum search algorithm [Grover 1996]. For convenience of the reader, we first briefly review the Grover algorithm, following Nielsen and Chuang [2000]. The search problem can be stated as follows. The search space consists of N elements, indexed by numbers $0, 1, \dots, N-1$. It is assumed that the search problem has exactly M solutions with $1 \leq M \leq \frac{N}{2}$, and we are supplied with an oracle, that is, a black box with the ability to recognize solutions to the search problem. Classically, it requires $O(\frac{N}{M})$ operations to find a solution. However, Grover [1996] discovered a quantum algorithm that requires only $O(\sqrt{\frac{N}{M}})$ operations. To present the Grover algorithm in an easier way, we may assume $N = 2^n$, so the index of the elements of the search space can be stored in n bits.

We do not distinguish an integer

$$t = \sum_{i=0}^{n-1} t_i 2^{n-1-i} \in \{0, 1, \dots, N-1\}$$

from its binary representation $t = t_0 t_1 \dots t_{n-1} \in \{0, 1\}^n$. So the characteristic function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ of the solutions is defined by

$$f(t) = \begin{cases} 1 & \text{if } t \text{ is a solution,} \\ 0 & \text{otherwise.} \end{cases}$$

The oracle can be represented by a unitary operator O on $n+1$ qubits:

$$|t\rangle|q\rangle \xrightarrow{O} |t\rangle|q \oplus f(t)\rangle$$

for all $t \in \{0, 1\}^n$ and $q \in \{0, 1\}$, where $\oplus$ is addition modulo 2, $|t\rangle$ is the index register, and $|q\rangle$ is the oracle qubit that is flipped if t is a solution, and is unchanged otherwise. It is easy to check that

$$|t\rangle|-\rangle \xrightarrow{O} (-1)^{f(t)}|t\rangle|-\rangle.$$

Observing the state of the $(n+1)$ th qubit remains $|-\rangle$ throughout the search algorithm, we can omit it for simplicity so that O can be seen as an n -qubit operator on the index register. To further simplify the presentation, we use the Grover operator G , which consists of the following steps:

- (1) Apply the oracle O ;
- (2) Apply the Hadamard transform $H^{\otimes n}$;
- (3) Perform a conditional phase shift $Ph: |0\rangle \rightarrow |0\rangle, |t\rangle \rightarrow -|t\rangle$ for all $t \neq 0$; that is, $Ph = 2|0\rangle\langle 0| - I$.
- (4) Apply the Hadamard transform $H^{\otimes n}$ again.

A routine calculation leads to $G = (2|\psi\rangle\langle\psi| - I)O$, where I is the identity operator on the index register of n qubits, and

$$|\psi\rangle = |+\rangle^{\otimes n} = \frac{1}{\sqrt{N}} \sum_{t=0}^{N-1} |t\rangle.$$

We write that

$$|\alpha\rangle = \frac{1}{\sqrt{N-M}} \sum_{t \text{ not solution}} |t\rangle,$$

$$|\beta\rangle = \frac{1}{\sqrt{M}} \sum_{t \text{ solution}} |t\rangle,$$

and $\cos \frac{\theta}{2} = \sqrt{\frac{N-M}{N}}$ ($0 \leq \frac{\theta}{2} \leq \frac{\pi}{2}$). Then it holds that $|\psi\rangle = \cos \frac{\theta}{2} |\alpha\rangle + \sin \frac{\theta}{2} |\beta\rangle$, $O|\alpha\rangle = |\alpha\rangle$, and $O|\beta\rangle = -|\beta\rangle$. A tedious calculation yields

$$G(\cos \delta |\alpha\rangle + \sin \delta |\beta\rangle) = \cos(\theta + \delta) |\alpha\rangle + \sin(\theta + \delta) |\beta\rangle. \quad (19)$$

Geometrically, the Grover operator G is a rotation: for all δ ,

$$(\cos \delta, \sin \delta) \xrightarrow{G} (\cos(\theta + \delta), \sin(\theta + \delta)),$$

in the two-dimensional space spanned by $|\alpha\rangle$ and $|\beta\rangle$, with the rotation angle θ . Now the Grover algorithm can be described as in Figure 4. The number R of iterations of

1. $|0\rangle^{\otimes n}|1\rangle$
2. $\xrightarrow{H^{\otimes(n+1)}} \frac{1}{\sqrt{2^n}} \sum_{t=0}^{2^n-1} |t\rangle|-\rangle = (\cos \frac{\theta}{2} |\alpha\rangle + \sin \frac{\theta}{2} |\beta\rangle)|-\rangle,$
3. $\xrightarrow{G^R} [\cos(\frac{2R+1}{2}\theta)|\alpha\rangle + \sin(\frac{2R+1}{2}\theta)|\beta\rangle]|-\rangle$
4. $\xrightarrow{\text{measure the first } n \text{ qubits in the computational basis}} |x\rangle|-\rangle$

Fig. 4. Grover search algorithm.

the Grover operator is a positive integer in the interval $[\frac{\pi}{2\theta} - 1, \frac{\pi}{2\theta}]$. We can see that the Grover algorithm requires only $O(\sqrt{\frac{N}{M}})$ operations in order to achieve a high success probability:

- (1) Since $0 \leq \frac{\theta}{2} \leq \frac{\pi}{2}$, we have $\frac{\theta}{2} \geq \sin \frac{\theta}{2} = \sqrt{\frac{M}{N}}$. This implies $R \leq \frac{\pi}{2\theta} \leq \frac{\pi}{4} \sqrt{\frac{N}{M}}$. So the algorithm requires $O(\sqrt{\frac{N}{M}})$ operations.
- (2) It holds that $\frac{\pi}{2} - \frac{\theta}{2} \leq \frac{2R+1}{2}\theta \leq \frac{\pi}{2} + \frac{\theta}{2}$. Thus, the success probability of the algorithms is

$$\begin{aligned} p_{succ} &= p(t \text{ is a solution}) = \sin^2 \left(\frac{2R+1}{2}\theta \right) \\ &\geq \cos^2 \frac{\theta}{2} = \frac{N-M}{N}. \end{aligned} \quad (20)$$

It is at least one-half because $M \leq \frac{N}{2}$. In particular, if $M \ll N$, then the success probability is very high.

We use quantum variables $q_0, q_1, \dots, q_{n-1}, q, r$, and their types are $type(q_i) = type(q) = \mathbf{Boolean}$ ($0 \leq i < n$) and $type(r) = \mathbf{integer}$. We write $\bar{q}$ for the quantum register consisting of $q_0, q_1, \dots, q_{n-1}$. The variable r is introduced to count the number of iterations of the Grover operator. Then the Grover algorithm can be written as a program *Grover* in the **while**-language defined in Section 3; see Figure 5.

The measurement $M = \{M_0, M_1\}$ in the loop is given as follows:

$$M_0 = \sum_{k \geq R} |k\rangle_r \langle k|, \quad M_1 = \sum_{k < R} |k\rangle_r \langle k|,$$

and the loop body D is given in Figure 6. In the measurement statement,

$$N = \{N_t = |t\rangle \langle t| : t \in \{0, 1\}^n\}$$

is the measurement in the computational basis of n qubits, and

$$\bar{S} = \{S_t \equiv \mathbf{Skip} : t \in \{0, 1\}^n\}.$$

```

 $q_0 := 0;$ 
 $q_1 := 0;$ 
.....
 $q_{n-1} := 0;$ 
 $q := 0;$ 
 $r := 0;$ 
 $q := \sigma_x q;$ 
 $q_0 := Hq_0;$ 
 $q_1 := Hq_1;$ 
.....
 $q_{n-1} := Hq_{n-1};$ 
 $q := Hq;$ 
while  $M[r] = 1$  do  $D;$ 
measure  $N[q_0, q_1, \dots, q_{n-1}] : \bar{S}$ 

```

Fig. 5. Quantum search program *Grover*.

```

 $q_0, q_1, \dots, q_{n-1} := O \ q_0, q_1, \dots, q_{n-1};$ 
 $q_0 := Hq_0;$ 
 $q_1 := Hq_1;$ 
.....
 $q_{n-1} := Hq_{n-1};$ 
 $q_0, q_1, \dots, q_{n-1} := Ph \ q_0, q_1, \dots, q_{n-1};$ 
 $q_0 := Hq_0;$ 
 $q_1 := Hq_1;$ 
.....
 $q_{n-1} := Hq_{n-1};$ 
 $r := r + 1$ 

```

Fig. 6. Loop body D .

The total correctness of the program *Grover* can be stated as follows: $\models_{tot} \{p_{succ} I\} Grover\{P\}$, where

$$I = \bigotimes_{i=0}^{n-1} I_{q_i} \otimes I_q \otimes I_r,$$

$$P = \left(\sum_{t \text{ solution}} |t\rangle_{\bar{q}} \langle t| \right) \otimes I_q \otimes I_r,$$

I_{q_i} ($t = 0, 1, \dots, n$) and I_q are the identity operators on $\mathcal{H}_2$; and I_r is the identity operator on $\mathcal{H}_\infty$. To avoid too complicated calculation, we choose to consider a very special case: $M = 1$ and $R = \frac{\pi}{2\theta} - \frac{1}{2}$ is the midpoint of the interval $[\frac{\pi}{2\theta} - 1, \frac{\pi}{2\theta}]$. In this case, there is a unique solution, say s , and $P = |s\rangle_{\bar{q}} \langle s| \otimes I_q \otimes I_r$. Also, we have $p_{succ} = 1$. So what we need to prove is then simply $\models_{tot} \{I\} Grover\{P\}$. By the soundness of qTD (Theorem 9.1), it suffices to show that $\vdash_{qTD} \{I\} Grover\{P\}$. We now prove it by using the proof rules given in Sections 8 and 9. The following simple lemma is useful.

LEMMA 9.1. *For each $i = 1, 2, \dots, n$, suppose that $\bar{q}_i$ is a quantum register and U_i is a unitary operator on $\mathcal{H}_{\bar{q}_i}$. Let $U = U_n \dots U_2 U_1$, where U_i stands for its cylinder extension according to Eq. (1) for every $i \leq n$. Then for any quantum predicate P , we have*

$$\vdash_{qPD} \{U^\dagger P U\} \bar{q}_1 := U_1 \bar{q}_1; \bar{q}_2 := U_2 \bar{q}_2; \dots; \bar{q}_n := U_n \bar{q}_n \{P\}.$$

PROOF. By repeatedly using (Axiom Unitary Transformation). $\square$

It is easy to see that

$$\sum_{t \in \{0, q\}^n} N_t^\dagger P N_t = P.$$

By (Axiom Skip) and (Rule Measurement) we obtain the following:

$$\vdash_{pTD} \{P\} \mathbf{measure} N[q_0, q_1, \dots, q_{n-1}] : \bar{S}\{P\}. \quad (21)$$

Put

$$P' = |s\rangle_{\bar{q}} \langle s| \otimes |-\rangle_q \langle -| \otimes |R\rangle_r \langle R|,$$

$$|\psi_k\rangle = \cos\left[\frac{\pi}{2} + (k - R)\theta\right] |\alpha\rangle + \sin\left[\frac{\pi}{2} + (k - R)\theta\right] |s\rangle$$

for every integer k , and

$$Q = \sum_{k < R} (|\psi_k\rangle_{\bar{q}} \langle \psi_k| \otimes |-\rangle_q \langle -| \otimes |k\rangle_r \langle k|).$$

We have that

$$\begin{aligned} M_0^\dagger P' M_0 + M_1^\dagger Q M_1 &= \sum_{k \leq R} (|\psi_k\rangle_{\bar{q}} \langle \psi_k| \otimes |-\rangle_q \langle -| \otimes |k\rangle_r \langle k|), \\ (G^\dagger \otimes I_q \otimes U_{+1}^\dagger) (M_0^\dagger P' M_0 + M_1^\dagger Q M_1) (G \otimes I_q \otimes U_{+1}) \\ &= \sum_{k \leq R} (|\psi_{k-1}\rangle_{\bar{q}} \langle \psi_{k-1}| \otimes |-\rangle_q \langle -| \otimes |k-1\rangle_r \langle k-1|) = Q. \end{aligned}$$

Thus, it follows from Lemma 9.1 and Eq. (19) that

$$\vdash_{qTD} \{Q\} D\{M_0^\dagger P' M_0 + M_1^\dagger Q M_1\}.$$

We define function $t : \mathcal{D}^-(\mathcal{H}_{\bar{q}} \otimes \mathcal{H}_q \otimes \mathcal{H}_r) \rightarrow \mathbb{N} \cup \{\infty\}$ as follows: if $\rho \in \mathcal{D}^-(\mathcal{H}_{\bar{q}} \otimes \mathcal{H}_q \otimes \mathcal{H}_r)$ can be written as

$$\rho = \sum_{k, l = -\infty}^{\infty} \rho_{kl} \otimes |k\rangle \langle l|$$

for some $\rho_{kl} \in \mathcal{D}^-(\mathcal{H}_{\bar{q}} \otimes \mathcal{H}_q)$ for all k, l , then $t(\rho) = \sup\{\max(k, l) | \rho_{kl} \neq 0\}$. Then

$$\begin{aligned} \llbracket D \rrbracket (M_1 \rho M_1^\dagger) &= \llbracket D \rrbracket \left(\sum_{k, l < R} \rho_{kl} \otimes |k\rangle_r \langle l| \right) \\ &= \sum_{k, l < R} [(G^\dagger \otimes I_q) \rho_{kl} (G \otimes I_q) \otimes |k-1\rangle_r \langle l-1|], \end{aligned}$$

and it follows that $t(\llbracket D \rrbracket (M_1 \rho M_1^\dagger)) < t(\rho)$. So t is a $(M_1^\dagger Q M_1, \epsilon)$ -bound function for any ϵ . By (Rule Loop Total) we assert that

$$\vdash_{qTD} \{M_0^\dagger P' M_0 + M_1^\dagger Q M_1\} \mathbf{while} M[r] = 1 \mathbf{do} D\{P'\}. \quad (22)$$

Using (Axiom Initialization) we obtain the following:

$$\left\{ \bigotimes_{i=0}^{k-1} |0\rangle_{q_i} \langle 0| \otimes \bigotimes_{i=k}^{n-1} I_{q_i} \otimes I_q \otimes I_r \right\} q_k := 0 = \left\{ \bigotimes_{i=0}^k |0\rangle_{q_i} \langle 0| \otimes \bigotimes_{i=k+1}^{n-1} I_{q_i} \otimes I_q \otimes I_r \right\}$$

for $k = 0, 1, \dots, n-1$, and they can be combined by (Rule Sequential Composition) to yield

$$\begin{aligned} \{I\} q_0 := 0; q_1 := 0; \dots; q_{n-1} := 0 & \left\{ \bigotimes_{i=0}^{n-1} |0\rangle_{q_i} \langle 0| \otimes I_q \otimes I_r \right\} \\ q &:= 0 \left\{ \bigotimes_{i=0}^{n-1} |0\rangle_{q_i} \langle 0| \otimes |0\rangle_q \langle 0| \otimes I_r \right\} \\ r &:= 0 \left\{ \bigotimes_{i=0}^{n-1} |0\rangle_{q_i} \langle 0| \otimes |0\rangle_q \langle 0| \otimes |0\rangle_r \langle 0| \right\} \\ q &:= \sigma_x q; \\ q_0 &:= Hq_0; q_1 := Hq_1; \dots; q_{n-1} := Hq_{n-1}; \\ q &:= Hq \{ |\psi\rangle_{\bar{q}} \langle \psi| \otimes |-\rangle_q \langle -| \otimes |0\rangle_r \langle -| \}. \end{aligned} \tag{23}$$

The last part of Eq. (23) is derived by Lemma 9.1 and

$$\begin{aligned} & [(H^\dagger)^\otimes \otimes \sigma_x^\dagger H^\dagger \otimes I_r] (|\psi\rangle_{\bar{q}} \langle \psi| \otimes |-\rangle_q \langle -| \otimes |0\rangle_r \langle 0|) (H^\otimes \otimes H\sigma_x \otimes I_r) \\ &= \bigotimes_{i=0}^{n-1} |0\rangle_{q_i} \langle 0| \otimes |0\rangle_q \langle 0| \otimes |0\rangle_r \langle 0|. \end{aligned}$$

It is obvious that $P' \subseteq P$. On the other hand, it follows from $R = \frac{\pi}{2\theta} - \frac{1}{2}$ that $|\psi\rangle = |\psi_0\rangle$. Then

$$\begin{aligned} |\psi\rangle_{\bar{q}} \langle \psi| \otimes |-\rangle_q \langle -| \otimes |0\rangle_r \langle -| &= |\psi_0\rangle_{\bar{q}} \langle \psi_0| \otimes |-\rangle_q \langle -| \otimes |0\rangle_r \langle -| \\ &\subseteq M_0^\dagger P' M_0 + M_1^\dagger Q M_1. \end{aligned}$$

We complete the proof by using (Rule Order) and (Rule Sequential Composition) to combine Eqs. (21), (22), and (23).

11. CONCLUSIONS

This article studied various the semantics of an imperative language for quantum programming, which is a quantum extension of the classical **while**-language, and presented a proof system for verifying correctness of quantum **while**-programs written in this language. The contribution of the article is twofold

- (1) The denotational and predicate transformer semantics of quantum **while**-programs have been defined in the previous literature; see for example [D'Hondt and Panangaden 2006]; [Feng et al. 2007]; [Selinger 2004b]; [Ying et al. 2010]. But this article chose to first define an operational semantics for quantum **while**-programs. Then their denotational and predicate transformer semantics can be derived from the operational semantics in an intrinsic and natural way.
- (2) Some program logics and proof rules have been introduced for the verification of quantum programs in the previous literature; see for example [Baltag and Smets 2004]; [Baltag and Smets 2006]; [Brunet and Jorrand 2004]; [Chadha et al. 2006];

[Feng et al. 2007]. In this article, however, a full-fledged Floyd–Hoare logic for deterministic quantum programs was developed for the first time, based on D’Hondt and Panangaden’s idea of representing quantum predicates by Hermitian operators and Selinger’s idea of modeling quantum programs by super-operators. In particular, its completeness, relative to the theory of the field of complex numbers, was proved. It is interesting to note that a technique from analysis (convergence of series) is employed in the completeness proof. This is quite different from the case for classical programs where only techniques from discrete mathematics are needed.

The combination of (1) and (2) allows us to put various semantics of and the proof system for quantum **while**-programs together to form a full picture.

An interesting problem for further studies is to find reasonable extensions of the quantum Floyd–Hoare logic presented in this article for bigger classes of quantum programs, including nondeterministic quantum programs [Zuliani 2004], parallel and distributed quantum programs. Hopefully, they will serve as a logical foundation for various effective techniques for verification of these bigger classes of quantum programs. But it is likely that certain essential difficulties will arise when we deal with parallel and distributed quantum programs due to the presence of entanglement in multipartite quantum systems, as experienced in the research of quantum process algebra [Gay and Nagarajan 2005]; [Jorrand and Lalire 2005]; [Feng et al. 2011]; [Ying et al. 2009].

APPENDIX: Proof of Proposition 5.3

(1) We proceed by induction on the structure of S .

Case 1. $S \equiv \mathbf{skip}$. Obvious.

Case 2. $S \equiv q := 0$. We only consider the case of $\text{type}(q) = \mathbf{integer}$, and the case of $\text{type}(q) = \mathbf{Boolean}$ is similar. It holds that

$$\begin{aligned} \text{tr}_{\text{var}(S)}(\llbracket S \rrbracket(\rho)) &= \sum_{n, n'=-\infty}^{\infty} q \langle n' | 0 \rangle_q \langle n | \rho | n \rangle_q \langle 0 | n' \rangle_q \\ &= \sum_{n=-\infty}^{\infty} q \langle n | \rho | n \rangle_q \\ &= \text{tr}_{\text{var}(S)}(\rho). \end{aligned}$$

Case 3. $S \equiv \bar{q} := U\bar{q}$. If $\{|\psi_i\rangle\}$ is an orthonormal basis of $\mathcal{H}_{\bar{q}}$, then $\{U^\dagger|\psi_i\rangle\}$ is also an orthonormal basis of $\mathcal{H}_{\bar{q}}$. Consequently,

$$\text{tr}_{\text{var}(S)}(\llbracket S \rrbracket(\rho)) = \sum_i \langle \psi_i | U \rho U^\dagger | \psi_i \rangle = \text{tr}_{\text{var}(S)}(\rho).$$

Case 4. $S \equiv S_1; S_2$. By the induction hypothesis on S_1 and S_2 , we have that

$$\begin{aligned} \text{tr}_{\text{var}(S)}(\llbracket S \rrbracket(\rho)) &= \text{tr}_{\text{var}(S_1) \cup \text{var}(S_2)}(\llbracket S_2 \rrbracket(\llbracket S_1 \rrbracket(\rho))) \\ &= \text{tr}_{\text{var}(S_1)}(\text{tr}_{\text{var}(S_2)}(\llbracket S_2 \rrbracket(\llbracket S_1 \rrbracket(\rho)))) \\ &= \text{tr}_{\text{var}(S_1)}(\text{tr}_{\text{var}(S_2)}(\llbracket S_1 \rrbracket(\rho))) \\ &= \text{tr}_{\text{var}(S_2)}(\text{tr}_{\text{var}(S_1)}(\llbracket S_1 \rrbracket(\rho))) \\ &= \text{tr}_{\text{var}(S_2)}(\text{tr}_{\text{var}(S_1)}(\rho)) \\ &= \text{tr}_{\text{var}(S)}(\rho). \end{aligned}$$

Case 5. $S \equiv \mathbf{measure} \ M[\bar{q}] : \bar{S}$. By the induction hypothesis on S_m for all outcomes m of measurement M , we obtain the following:

$$\begin{aligned}
 tr_{var(S)}(\llbracket S \rrbracket(\rho)) &= \sum_m tr_{var(S)}(\llbracket S_m \rrbracket(M_m \rho M_m^\dagger)) \\
 &= \sum_m tr_{var(S) \setminus var(S_m)}(tr_{var(S_m)}(\llbracket S_m \rrbracket(M_m \rho M_m^\dagger))) \\
 &= \sum_m tr_{var(S) \setminus var(S_m)}(tr_{var(S_m)}(M_m \rho M_m^\dagger)) \\
 &= \sum_m tr_{var(S)}(M_m \rho M_m^\dagger) \\
 &= \sum_m tr_{var(S) \setminus \{\bar{q}\}}(tr_{\bar{q}}(M_m \rho M_m^\dagger)) \\
 &= \sum_m tr_{var(S) \setminus \{\bar{q}\}}(tr_{\bar{q}}(M_m^\dagger M_m \rho)) \\
 &= tr_{var(S) \setminus \{\bar{q}\}} \left(tr_{\bar{q}} \left(\left(\sum_m M_m^\dagger M_m \right) \rho \right) \right) \\
 &= tr_{var(S) \setminus \{\bar{q}\}}(tr_{\bar{q}}(\rho)) \\
 &= tr_{var(S)}(\rho)
 \end{aligned}$$

because measurement M satisfies the normalization condition $\sum_m M_m^\dagger M_m = I_{\mathcal{H}_{\bar{q}}}$.

Case 6. $S \equiv \mathbf{while} \ M[\bar{q}] = 1 \ \mathbf{do} \ S'$. For simplicity, we write **while** for quantum loop “**while** $M[\bar{q}] = 1$ **do** S' ”. First, we have

$$\begin{aligned}
 tr_{var(S)}(\llbracket (\mathbf{while})^0 \rrbracket(\rho)) &= 0_{\mathcal{H}_{all}}, \\
 tr_{var(S)}(\llbracket (\mathbf{while})^n \rrbracket(\rho)) &= tr_{var(S)}(\rho)
 \end{aligned} \tag{24}$$

for all $n \geq 1$. This can be proved by induction on n . In fact,

$$(\mathbf{while})^{k+1} \equiv \mathbf{measure} \ M[\bar{q}] : \bar{S}$$

where $\bar{S} \equiv S_0, S_1, S_0 \equiv \mathbf{skip}$, and $S_1 \equiv S'$; $(\mathbf{while})^k$. Thus, by Cases 1, 4, and 5, we can immediately derive Eq. (24) for $n = k + 1$ from the induction hypothesis for $n = k$. Now, it follows from continuity of trace that

$$\begin{aligned}
 tr_{var(S)}(\llbracket S \rrbracket(\rho)) &= tr_{var(S)} \left(\bigvee_{n=0}^{\infty} \llbracket (\mathbf{while})^n \rrbracket(\rho) \right) \\
 &= \bigvee_{n=0}^{\infty} tr_{var(S)}(\llbracket (\mathbf{while})^n \rrbracket(\rho)) \\
 &= tr_{var(S)}(\rho).
 \end{aligned}$$

(2) The trick is to prove the following slightly stronger conclusion:

—*Claim:* For any $var(S) \subseteq X \subseteq var$, $tr_{X \setminus var(S)}(\rho_1) = tr_{X \setminus var(S)}(\rho_2)$ implies $tr_{X \setminus var(S)}(\llbracket S \rrbracket(\rho_1)) = tr_{X \setminus var(S)}(\llbracket S \rrbracket(\rho_2))$.

We also proceed by induction on the structure of S .

Case 1. $S \equiv \mathbf{skip}$. Obvious.

Case 2. $S \equiv q := 0$. We only consider the case of $\text{type}(q) = \mathbf{integer}$, the case of $\text{type}(q) = \mathbf{Boolean}$ is similar. First, let $\{|\psi_i\rangle\}$ be an orthonormal basis of $\mathcal{H}_{X \setminus \{q\}}$, then we have

$$\begin{aligned}
 \text{tr}_{X \setminus \text{var}(S)}(\llbracket S \rrbracket(\rho)) &= \text{tr}_{X \setminus \{q\}} \left[\sum_{n=-\infty}^{\infty} (|0\rangle_q \langle n| \otimes I_{\mathcal{H}_{\text{Var} \setminus \{q\}}}) \rho (|n\rangle_q \langle 0| \otimes I_{\mathcal{H}_{\text{Var} \setminus \{q\}}}) \right] \\
 &= \sum_i \sum_{n=-\infty}^{\infty} (I_{\mathcal{H}_{(\text{Var} \setminus X) \cup \{q\}}} \otimes \langle \psi_i |) (|0\rangle_q \langle n| \otimes I_{\mathcal{H}_{\text{Var} \setminus \{q\}}}) \rho \\
 &\quad (|n\rangle_q \langle 0| \otimes I_{\mathcal{H}_{\text{Var} \setminus \{q\}}}) (I_{\mathcal{H}_{(\text{Var} \setminus X) \cup \{q\}}} \otimes |\psi_i\rangle) \\
 &= \sum_{n=-\infty}^{\infty} ((|0\rangle_q \langle n| \otimes I_{\mathcal{H}_{\text{Var} \setminus \{q\}}}) \sum_i (I_{\mathcal{H}_{(\text{Var} \setminus X) \cup \{q\}}} \otimes \langle \psi_i |) \rho \\
 &\quad (I_{\mathcal{H}_{(\text{Var} \setminus X) \cup \{q\}}} \otimes |\psi_i\rangle) (|n\rangle_q \langle 0| \otimes I_{\mathcal{H}_{\text{Var} \setminus \{q\}}})) \\
 &= \sum_{n=-\infty}^{\infty} |0\rangle_q \langle n| \text{tr}_{X \setminus \text{var}(S)}(\rho) |n\rangle_q \langle 0|.
 \end{aligned}$$

Thus, it is easy to see that (2) holds in this case.

Case 3. $S \equiv \bar{q} := U\bar{q}$. Assume that $\{|\psi_i\rangle\}$ is an orthonormal basis of $\mathcal{H}_{X \setminus \{\bar{q}\}}$. Then it holds that

$$\begin{aligned}
 \text{tr}_{X \setminus \text{var}(S)}(\llbracket S \rrbracket(\rho)) &= \text{tr}_{X \setminus \{\bar{q}\}} (U \otimes I_{\mathcal{H}_{\text{Var} \setminus \{\bar{q}\}}}) \rho (U^\dagger \otimes I_{\mathcal{H}_{\text{Var} \setminus \{\bar{q}\}}}) \\
 &= \sum_i (I_{\mathcal{H}_{(\text{Var} \setminus X) \cup \{\bar{q}\}}} \otimes \langle \psi_i |) (U \otimes I_{\mathcal{H}_{\text{Var} \setminus \{\bar{q}\}}}) \rho (U^\dagger \otimes I_{\mathcal{H}_{\text{Var} \setminus \{\bar{q}\}}}) (I_{\mathcal{H}_{(\text{Var} \setminus X) \cup \{\bar{q}\}}} \otimes |\psi_i\rangle) \\
 &= (U \otimes I_{\mathcal{H}_{\text{Var} \setminus \{\bar{q}\}}}) \sum_i (I_{\mathcal{H}_{(\text{Var} \setminus X) \cup \{\bar{q}\}}} \otimes \langle \psi_i |) \rho (I_{\mathcal{H}_{(\text{Var} \setminus X) \cup \{\bar{q}\}}} \otimes |\psi_i\rangle) (U^\dagger \otimes I_{\mathcal{H}_{\text{Var} \setminus \{\bar{q}\}}}) \\
 &= (U \otimes I_{\mathcal{H}_{\text{Var} \setminus \{\bar{q}\}}}) \text{tr}_{X \setminus \text{var}(S)}(\rho) (U^\dagger \otimes I_{\mathcal{H}_{\text{Var} \setminus \{\bar{q}\}}})
 \end{aligned}$$

because $|\psi_i\rangle \in \mathcal{H}_{X \setminus \{\bar{q}\}}$, $U \in \mathcal{L}(\mathcal{H}_{\bar{q}})$, and $(X \setminus \{\bar{q}\}) \cap \{\bar{q}\} = \emptyset$. Now it is easy to see that (2) holds in this case.

Case 4. $S \equiv S_1; S_2$. If $\text{var}(S_1) \cup \text{var}(S_2) = \text{var}(S) \subseteq X$, then there are $Y, Z \subseteq \text{Var}$ such that $\text{var}(S_1) \subseteq Y$, $\text{var}(S_2) \subseteq Z$ and

$$Y \setminus \text{var}(S_1) = X \setminus \text{var}(S) = Z \setminus \text{var}(S_2).$$

Thus, we have

$$\begin{aligned}
 \text{tr}_{Y \setminus \text{var}(S_1)}(\rho_1) &= \text{tr}_{X \setminus \text{var}(S)}(\rho_1) \\
 &= \text{tr}_{X \setminus \text{var}(S)}(\rho_2) \\
 &= \text{tr}_{Z \setminus \text{var}(S_2)}(\rho_2).
 \end{aligned}$$

By the induction hypothesis on S_1 , we obtain the following:

$$\begin{aligned}
 \text{tr}_{Z \setminus \text{var}(S_2)}(\llbracket S_1 \rrbracket(\rho_1)) &= \text{tr}_{Y \setminus \text{var}(S_1)}(\llbracket S_1 \rrbracket(\rho_1)) \\
 &= \text{tr}_{Y \setminus \text{var}(S_1)}(\llbracket S_1 \rrbracket(\rho_2)) \\
 &= \text{tr}_{Z \setminus \text{var}(S_2)}(\llbracket S_1 \rrbracket(\rho_2)).
 \end{aligned}$$

Furthermore, by the induction hypothesis on S_2 , we obtain

$$\begin{aligned} \text{tr}_{X \setminus \text{var}(S)}(\llbracket S \rrbracket(\rho_1)) &= \text{tr}_{Z \setminus \text{var}(S_2)}(\llbracket S_2 \rrbracket(\llbracket S_1 \rrbracket(\rho_1))) \\ &= \text{tr}_{Z \setminus \text{var}(S_2)}(\llbracket S_2 \rrbracket(\llbracket S_1 \rrbracket(\rho_2))) \\ &= \text{tr}_{X \setminus \text{var}(S)}(\llbracket S \rrbracket(\rho_2)). \end{aligned}$$

Case 5. $S \equiv \mathbf{measure} \ M[\bar{q}] : \bar{S}$. First, let $\{|\psi_i\rangle\}$ be an orthonormal basis of $\mathcal{H}_{X \setminus \text{var}(S)}$. Then, for any outcome m of measurement M , we have the following:

$$\begin{aligned} \text{tr}_{X \setminus \text{var}(S)}(M_m \rho M_m^\dagger) &= \sum_i (I_{\mathcal{H}_{\text{Var}(X \setminus \text{var}(S))}} \otimes \langle \psi_i |) (M_m \otimes I_{\mathcal{H}_{\text{Var}(\bar{q})}}) \rho \\ &\quad (M_m^\dagger \otimes I_{\mathcal{H}_{\text{Var}(\bar{q})}}) (I_{\mathcal{H}_{\text{Var}(X \setminus \text{var}(S))}} \otimes |\psi_i\rangle) \\ &= (M_m \otimes I_{\mathcal{H}_{\text{Var}(\bar{q})}}) \sum_i (I_{\mathcal{H}_{\text{Var}(X \setminus \text{var}(S))}} \otimes \langle \psi_i |) \rho \\ &\quad (I_{\mathcal{H}_{\text{Var}(X \setminus \text{var}(S))}} \otimes |\psi_i\rangle) (M_m^\dagger \otimes I_{\mathcal{H}_{\text{Var}(\bar{q})}}) \\ &= M_m \text{tr}_{X \setminus \text{var}(S)}(\rho) M_m^\dagger \end{aligned}$$

because $|\psi_i\rangle \in \mathcal{H}_{X \setminus \text{var}(S)}$, $M_m \in \mathcal{L}(\mathcal{H}_{\bar{q}})$, and $(X \setminus \text{var}(S)) \cap \{\bar{q}\} = \emptyset$. Consequently, it follows from $\text{tr}_{X \setminus \text{var}(S)}(\rho_1) = \text{tr}_{X \setminus \text{var}(S)}(\rho_2)$ that

$$\text{tr}_{X \setminus \text{var}(S)}(M_m \rho_1 M_m^\dagger) = \text{tr}_{X \setminus \text{var}(S)}(M_m \rho_2 M_m^\dagger).$$

Note that we can write $X \setminus \text{var}(S) = Y \setminus \text{var}(S_m)$ for some $Y \supseteq \text{var}(S_m)$. Then it holds that

$$\text{tr}_{Y \setminus \text{var}(S_m)}(M_m \rho_1 M_m^\dagger) = \text{tr}_{Y \setminus \text{var}(S_m)}(M_m \rho_2 M_m^\dagger).$$

By the induction hypothesis on S_m , we obtain the following:

$$\begin{aligned} \text{tr}_{X \setminus \text{var}(S)}(\llbracket S_m \rrbracket(M_m \rho_1 M_m^\dagger)) &= \text{tr}_{Y \setminus \text{var}(S_m)}(\llbracket S_m \rrbracket(M_m \rho_1 M_m^\dagger)) \\ &= \text{tr}_{Y \setminus \text{var}(S_m)}(\llbracket S_m \rrbracket(M_m \rho_2 M_m^\dagger)) \\ &= \text{tr}_{X \setminus \text{var}(S)}(\llbracket S_m \rrbracket(M_m \rho_2 M_m^\dagger)). \end{aligned}$$

Therefore, it follows that

$$\begin{aligned} \text{tr}_{X \setminus \text{var}(S)}(\llbracket S \rrbracket(\rho_1)) &= \text{tr}_{X \setminus \text{var}(S)} \sum_m \llbracket S_m \rrbracket(M_m \rho_1 M_m^\dagger) \\ &= \sum_m \text{tr}_{X \setminus \text{var}(S)}(\llbracket S_m \rrbracket(M_m \rho_1 M_m^\dagger)) \\ &= \sum_m \text{tr}_{X \setminus \text{var}(S)}(\llbracket S_m \rrbracket(M_m \rho_2 M_m^\dagger)) \\ &= \text{tr}_{X \setminus \text{var}(S)} \sum_m \llbracket S_m \rrbracket(M_m \rho_2 M_m^\dagger) \\ &= \text{tr}_{X \setminus \text{var}(S)}(\llbracket S \rrbracket(\rho_2)). \end{aligned}$$

Case 6. $S \equiv \mathbf{while} \ M[\bar{q}] = 1 \ \mathbf{do} \ S'$. Assume that $\text{tr}_{X \setminus \text{var}(S)}(\rho_1) = \text{tr}_{X \setminus \text{var}(S)}(\rho_2)$. Using Cases 1, 4, and 5, we can show that

$$\text{tr}_{X \setminus \text{var}(S)}(\llbracket (\mathbf{while})^n \rrbracket(\rho_1)) = \text{tr}_{X \setminus \text{var}(S)}(\llbracket (\mathbf{while})^n \rrbracket(\rho_2))$$

for all $n \geq 0$ by induction on n . Therefore, we obtain

$$\begin{aligned}
 \text{tr}_{X \setminus \text{var}(S)}(\llbracket \mathbf{while} \rrbracket(\rho_1)) &= \text{tr}_{X \setminus \text{var}(S)}\left(\bigvee_{n=0}^{\infty} \llbracket (\mathbf{while})^n \rrbracket(\rho_1)\right) \\
 &= \bigvee_{n=0}^{\infty} \text{tr}_{X \setminus \text{var}(S)}(\llbracket (\mathbf{while})^n \rrbracket(\rho_1)) \\
 &= \bigvee_{n=0}^{\infty} \text{tr}_{X \setminus \text{var}(S)}(\llbracket (\mathbf{while})^n \rrbracket(\rho_2)) \\
 &= \text{tr}_{X \setminus \text{var}(S)}\left(\bigvee_{n=0}^{\infty} \llbracket (\mathbf{while})^n \rrbracket(\rho_2)\right) \\
 &= \text{tr}_{X \setminus \text{var}(S)}(\llbracket \mathbf{while} \rrbracket(\rho_2))
 \end{aligned}$$

by continuity of trace.

ACKNOWLEDGMENTS

The author is very grateful to the referees for their useful comments and instructive suggestions.

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Received September 2010; revised March 2011; accepted July 2011